



Machine learning and deep learning based psoriasis recognition system: evaluation, management, prognosis—where we are and the way to the future

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Abstract

Psoriasis, a chronic inflammatory skin disease, poses significant diagnostic challenges due to its heterogeneous clinical manifestations and overlap with other dermatological conditions. The inconsistencies in diagnostic standards and variability in disease presentation further exacerbate the controversy surrounding the diagnosis of psoriasis. Recent advances in artificial intelligence (AI), particularly deep learning (DL) techniques such as convolutional neural networks (CNNs) and transformers, have shown promising results in improving recognition of psoriasis. This comprehensive review aims to provide an in-depth examination of the current state of AI applications in the diagnosis of psoriasis, highlighting the strengths and limitations of existing approaches. We evaluate the performance of various AI models on publicly available datasets, including the Psoriasis Image Dataset (PID) and the International Skin Imaging Collaboration (ISIC) dataset, and compare their accuracy, sensitivity, and specificity with traditional diagnostic methods. Furthermore, we discuss the challenges and future directions of AI-powered psoriasis diagnosis, including the need for large, diverse datasets, addressing data quality and generalization issues, and developing explainable AI models. Our review highlights the potential of AI to revolutionize the diagnosis and management of psoriasis and provides a pathway for future research in this field.

Keywords Psoriasis · Eczema · Seborrheic dermatitis (seborrhea) · Psoriasis Image Dataset (PID) · International Skin Imaging Collaboration (ISIC) · Artificial intelligence · Machine learning · Deep learning

1 Introduction

The skin plays a vital role in controlling the body's temperature and defends against viruses, germs, allergies, bacterial, and fungal infections (Ahmad Fadzil et al. 2009). However, many people endure skin disorders due to innumerable causes like internal causes (genetics, body glands, hormones, specific conditions, acne, etc.) and external causes (air pollution, rashes,

heat burn, scalding, sensitivity to sunlight, etc.). Contagious skin conditions include folliculitis, scabies, cold sores, ringworm, and lice, whereas non-contagious conditions include acne, rosacea, and medication allergies. Chronic conditions include psoriasis, sweet syndrome, and fuji disease. According to global statistics, 1.79% of physical disability among people is caused by skin diseases (Hameed et al. 2020). In many nations across the world, between 30% and 70% of the population are affected by skin disorders (Yang et al. 2018; Saba et al. 2019; Ali et al. 2021).

Skin is the largest organ in the body that covers whole external surface of the body. Skin is made up of three layers: Epidermis, Dermis, and Hypodermis or Subcutis layer. Skin layers are affected by different skin diseases. For example, Epidermis is affected by Acne, Boils, Basal cell carcinoma, Dandruff, Eczema, Melanoma, Melasma, Actinic, keratosis, Squamous cell carcinoma, Merkel cell carcinoma, and Psoriasis (Nawaz et al. 2022; Khan et al. 2021). Ehlers-Danlos syndrome, Solar elastosis, Hives (urticaria), Primary dermal melanoma, Marfan syndrome, Systemic sclerosis (scleroderma) and skin Tumors affects the Dermis layer of the skin. Bedsores, Hypothermia, Sarcoidosis, Panniculitis, and Third-degree burns affects the Hypodermis or Subcutis layer. Fig 1 depicts the layers of the skin that indicate normal skin and psoriasis skin.

Psoriasis is an insistent, relapsing, chronic, and inflammatory skin disorder with no treatment and cannot be passed from one individual to another. According to the IFPA (International Federation of Psoriasis Associations) report, approximately 125 million people are affected by psoriasis globally and 3 to 5% of Pakistani people are suffering from psoriasis (George et al. 2016). When the immune system sends faulty signals that lead to the excessive creation of new skin cells and affect normal skin cells for the pathogen, it develops psoriasis. Psoriasis is an autoimmune disorder that causes red scabby patches on the skin. Psoriasis produces patchy skin about 10 times faster than normal skin. Psoriasis seems to be small red bumps that grow and scale. These scales are stuck and can be shed easily, but they may be torn down if scratched. The rash which is formed by scratching will be transformed into lesions. The lesion will damage the larger area of the skin.

Psoriasis is generally caused by the disturbance of the immune system. Our immune system defends us from certain diseases and fights against infections. The immune system use T-cells which is one of the main types of cell to fight against infection, bacteria, and

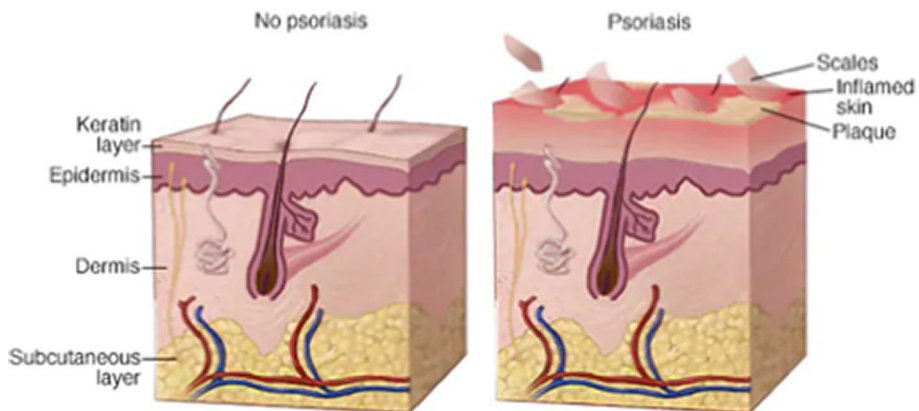


Fig. 1 Skin layers indicating psoriasis and no psoriasis

germs. Due to other overreactions of the immune system, new cells grow rapidly on the skin (almost 3–4 days). These new cells will be replaced with the older ones and then silvery scales of psoriasis are formed. There may be genetic components that pass from parents to children. Common psoriasis symptoms may include streptococcal or other infection that affects the immune system, cold weather, skin injuries, and certain prescription medications.

Psoriasis can affect people of any age but most commonly affect adults and women. Psoriasis symptoms vary from person to person. Some persons may only have rare patches of the affected skin, while others may have prevalent psoriasis covering the large skin area. The most common symptoms are red scaly skin patches, thick plaques with white or silvery scales, itching, burning, cracked skin, swollen, and painful joints, etc.

Psoriasis can be diagnosed by examining the outer surface of the skin using clinical features. Other skin disorders like Eczema, Seborrheic dermatitis (seborrhea), Pityriasis, Tinea corporis, and Lichen planus have the same clinical features as psoriasis (Kim et al. 2015). Therefore, the diagnosis of psoriasis had been a controversial issue for last decades. However, several medical and computational treatments like Light therapy, Methotrexate, Retinoids, Calcineurin inhibitors, Psychotherapy, and Phototherapy are used for the diagnosis of psoriasis. Computer-aided techniques play an important role in the detection, classification, and diagnosis of psoriasis. Convolutional Neural Network (CNN) has been introduced in recent years for the detection and classification of dermoscopic images. The CNN uses convolutional filters or kernels to detect the pixel-level descriptive features from a large data set of dermoscopic images and the corresponding diagnosis as input for "end-to-end" training. The strength of CNN lies in its ability to learn "what to look for" without manual guidance. Since high-resolution information is attained by dermoscopic images and the appearance of the skin lesion mainly results into the diagnosis of psoriasis. A CNN trained on dermoscopic images would achieve excellent performance in identifying psoriasis, which might improve the management of psoriasis in areas with limited medical resources (Bibi et al. 2023; Dillshad et al. 2023). Most of the current research is focused on the field of neoplastic skin diseases, such as skin cancer, rather than pigmented skin diseases, and there are fewer studies on pigmented skin diseases. Therefore, current studies about the computer-aided diagnosis of skin diseases focus mainly on the classification of psoriasis. With the emergence of Artificial Intelligence techniques like deep learning and machine learning, a tremendous improvement in the psoriasis recognition system can be observed. The survey paper aims to review, evaluate, and synthesize the concepts of artificial intelligence in psoriasis recognition systems, examining both its current applications and its future potential.

The rest of the paper is organized as: Sect. 2 presents the background details of psoriasis according to the biological point of view. Furthermore, Sect. 2 also presents types of psoriasis, pathogens, affected areas, and risk factors for the development of psoriasis. Section 3 discuss the overview of artificial intelligence, machine learning, and deep learning. Section 4 explains and depicts the Research Methodology of the survey. The details about datasets used in the domain of Psoriasis are elaborated in Sect. 5. The state of art according to the taxonomy of deep learning based studies and conventional machine learning are presented in Sect. 6. Section 7 discuss the current challenges, limitations, trends and future outlooks in the domain of Psoriasis detection. Section 8 presents the Evaluation, Synthesis, and Main Findings of the review. Finally Sect. 9 presents the conclusion of the review.

2 Background

This section presents the detailed medical background of psoriasis. Psoriasis is an autoimmune disorder that causes red scabby patches on the skin. Psoriasis produces patchy skin about 10 times faster than normal skin. Generally, skin cells are replaced during 10 to 30 days but when psoriasis grows, new cells are reproduced after 3 to 4 days. Thus, detaching of old skin cells and the creation of new skin results in the formation of red skin patches with silvery scales (Razzak et al. 2020; Xing et al. 2024). Psoriasis can produce anywhere, but mostly seems on the knees, elbows, scalp, and lower back. Common psoriasis symptoms may include streptococcal or other infection that affects the immune system, cold weather, skin injuries, and certain prescription medications. The key risk factors of psoriasis are environmental factors and genetic predisposition (Aijaz et al. 2022; Liu et al. 2024). Based on the symptoms, affected areas, factors, and characteristics psoriasis can be categorized into different types. This section provides comprehensive details about the types of psoriasis, pathogens of psoriasis, affected areas of psoriasis, and risk factors for the development of psoriasis.

2.1 Types of psoriasis

There are several types of psoriasis depending on the characteristics: Inverse, Pustular, Erythrodermic Plaque and Guttate. Kaddu et al. (1999) as depicted in Fig 2. *Inverse psoriasis* also known as flexural or intertriginous psoriasis affects where the two skin surface folds. About 3 to 7% of patients with psoriasis suffer from inverse psoriasis. It seems to be common psoriasis, except for only two reasons i.e. the area of the lesions and their appearance. The irritation in inverse psoriasis may be increased due to the sweating and rubbing on the location where plaques occur. The most commonly affected area is the inguinal fold, which is followed by the external genitalia and axilla. Inverse psoriasis is mostly found on the extensor surfaces of the body and trunk, such as the sacral area, knees, scalp, and elbows. Treatments include a range of systematic and topical therapies along with phototherapy. The National Psoriasis Foundation recommended the treatment of inverse psoriasis, including immunomodulators and calcipotriene, and the use of a minimal dose of topical corticosteroids for acute exacerbations for long-term therapy (Reynolds et al. 2019).

Plaque psoriasis, also referred to as *Vulgaris psoriasis*, represents approximately 90% of psoriatic patients. Plaque psoriasis tends to appear in the form of red and thick patches

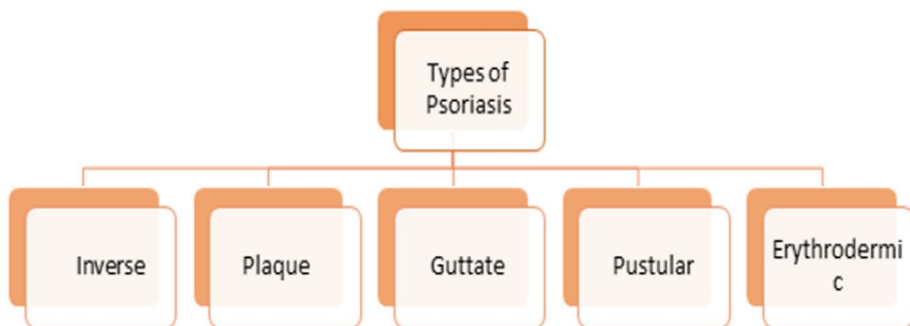


Fig. 2 Taxonomy depicting the types of psoriasis

on the skin surface, usually covered by white silvery scales, psoriasis appears on the knees, elbows, and scalp. The well-known features of plaques are psoriasis Auspitz phenomenon and the Koebner phenomenon. Topical treatments include salicylic acid, vitamin D products, topical corticosteroids, vitamin A products, dithranol, and tar-based preparations (Chalmers et al. 2001).

Guttate psoriasis, a type of psoriasis that develop on the skin as small, red, scaly, patches. About one-third of people come with this disease. Guttate psoriasis is not as common as plaque psoriasis. The spots of guttate psoriasis are thick as compared to plaque psoriasis, but it does not leave a scar. These spots or patches are named teardrops. The pink-red, scaly patches will spread on the face, ears, and scalp. They also show up on arms, legs, palms, the soles of your feet, or nails and chest (Chalmers et al. 2001). About 25% cases of guttate psoriasis come with ripened chronic plaque psoriasis. Two major factors associated with guttate psoriasis i.e. Genetics and Bacterial infections. In the case of genetics, it comes from parents or family members and in the case of bacterial infections, guttate psoriasis typically develops within 1 or 2 weeks after the streptococcal infection of tonsillitis and upper respiratory tract. People with autoimmune disorders, HIV, or whose immune systems are suppressed by chemotherapy have a higher risk of affecting guttate psoriasis. Guttate psoriasis can be diagnosed by skin biopsy, throat swap culture, and blood tests. Treatments for guttate psoriasis depend on the severity of the patient's situation. Lotions and creams that reduce inflammation are used for milder cases. Whereas severe cases may be treated as biological medications and phototherapy (Kaddu et al. 1999).

Pustular psoriasis, is another severe form of psoriasis that causes inflammation on the skin with yellow and white blisters filled with pus. The pus is composed of white blood cells. These blisters can be painful and never will be a sign of bacteria or infection (Kaddu et al. 1999). This type of psoriasis occurs because of changes occurring in the immune system in the way the cells work. The crusty pustules or patches that appear on the skin's light skin will appear red while on darker skin they will appear as a dark scar (Hoegler et al. 2018). According to Dermnetz NZ, pustules that appear on the skin are about 2–3 mms in diameter. This type of psoriasis is not contagious as others. It also adversely affects women during pregnancy. According to a dermatologist, pustular psoriasis can be diagnosed and identified from a blood count, a skin biopsy, or liver and kidney function tests. Depending on the symptoms that appear on the skin, pustular psoriasis has several types. Palmoplantar Pustular Psoriasis is the type where the symptoms appear on the soles of feet, palms, and inside the heels and thumbs. Acrodermatitis continua of Hallopeau (ACH) is when symptoms appear on toes, nails, and particularly on the tips of the fingers (Robinson et al. 2012).

Erythrodermic psoriasis, is a common type of psoriasis in which the red rashes spread over the whole body in a couple of days. The skin rash is very irritated which cause a burning sensation and may results into peeling of the skin. The rash also affects the skin to regulate our body temperature. Plaque psoriasis is present in around one third of those who develop erythrodermic psoriasis. Some people develop erythrodermic psoriasis after having stress, infection or illness, allergic reaction to the medicine, or severe sunburn. Erythrodermic psoriasis treatments include biological therapies, oral medications, and topical steroid applications until symptoms have improved (Rosenbach et al. 2010).

2.2 Pathogenesis of psoriasis

Psoriasis is a skin disease that causes hyperproliferation and rapid epidermal changes. The pathogenesis of psoriasis plays an important role in various cellular mechanisms and the role of T cells, macrophages, keratinocytes, antigen-presenting cells (APCs), Langerhans cells, natural killer cells, keratinocytes growth factor (KGF), an array of Th1?type cytokines, and vascular endothelial growth factor (VEGF), etc. (McGregor et al. 1992). Psoriasis is related to the immune system's disease in which T lymphocyte activation results in epidermal hyperproliferation and dermatological inflammation. The pathogens of psoriasis involve various processes (Rajguru et al. 2020) like:

- The activation of T-cells
 - Angiogenesis
 - Genetic factors
 - Role of dendritic cell
 - Hyperproliferation of keratinocytes
- T-Cell activation* is the common pathogen of psoriasis. The origination and development of disease occur when abnormal T-cells combined with the relation between complex cytokine network and keratinocytes (Ortonne 1999). Any chemical or physiological injury caused by the leading flaw in keratinocytes could trigger secretion and synthesis of cytokines which eventually leads to the antigen-independent activation of T lymphocytes. However, this situation results in the secretion of additional cytokines as well as an increase in the set of keratinocytes, T lymphocytes, and inflammation. In comparison to cytokines secreted by healthy epidermal cells, psoriatic epidermal cells produce more cytokines that cause T lymphocyte stimulation. Bos and De Rie (1999). Additionally, it was proposed that only psoriatic keratinocytes exhibit overexcited propagation in response to activated T-cell signals because they possess particular receptors or signal-transducing mechanisms. Chang et al. (1992). Psoriatic T cell supernatants do not cause an adverse reaction in abnormal keratinocytes. Bata-Csorgo et al. (1995).

The mechanism of Angiogenesis in psoriasis is still unidentified, however, the most important cause of pro-angiogenic cytokines (VEGF, IL-8) is keratinocytes. In spite of that, it is clear in emerging psoriatic plaque that endothelial cells display Weibel-Palade bodies and prominent Golgi apparatus due to the swelling and activating (Christophers and Mrowietz 1995). After activation, By migrating, sprouting, and laying down a BM containing pericytes for structural maintainance, these cells develop new vessel networks. Longo et al. (2002). Dermal blood vessels dilate and intercellular spaces of endothelial cells become spread. The lesion capillary loops espouse a venous phenotype that involves joined express E-selection, fenestrations, and migration of leukocytes into the skin. Even though psoriasis' etiology involves angiogenesis, understanding the mechanisms that lead to angio-proliferation may aid in the development of novel antipsoriatic medications. Nickoloff (2000). However, Retinoids, cyclosporine vitamin D, and analogs contain anti-angiogenic activity along with anti-inflammatory and antiproliferative effects (Griffioen and Molema 2000). According to researchers, psoriasis is one of the intricate diseases in which the environment and inheritance are multifactorial - genetic variants in multiple genes act together. Disease susceptibility location have been suggested as factors of predisposition (Trembath et al. 1997).

The growth of many cell types is held responsible for increased proliferation. In the case of *hyperproliferating psoriatic keratinocytes*, the time of the cell cycle is short and it takes only 4 days in the psoriatic epidermis. However, in normal epidermis, it takes 26 days for maturing and peeling of keratinocytes (Gottlieb 2003). Antipsoriatic drugs affect keratinocyte proliferation in which Calcipotriol, the synthetic and natural vitamin A derivatives and a vitamin D3 analog and retinoids, reduce variation and keratinocyte hyperproliferation. On human epidermal keratinocytes, cyclosporine exhibits potent immunomodulatory and antiproliferative actions. Krueger and Bowcock (2005).

2.3 Affected areas of psoriasis

Psoriasis causes significant stigma for patients and has a significant economic impact. Skin involvement can affect specific localized areas or the whole body. Psoriasis phenotypes, according to the involvement of anatomical location are nail, scalp, and intertriginous areas (Gottfried 2008).

The *Scalp psoriasis* is considered the most common site of disease involvement. According to the psoriasis report of north Western Europe, there are approximately 50% to 80% of psoriasis patients arise with scalp psoriasis, and 1.5% to 2% of patients' bodies lead to the prevalence of scalp psoriasis. Psoriasis affects more than half of the scalp (van de Kerkhof and Franssen 2001). Dutch questionnaire revealed that 81% of patients had scalp psoriasis for more than 5 years. The erythemasquamous lesions on the scalp are characterized by silver-white scaling. Beyond the neck, face, and retro auricular regions are the places where the psoriatic lesions extend. The first sign of psoriasis in young adults and children is the appearance of crusted plaque on the whole scalp (Kim et al. 2017). Generally, scalp psoriasis causes loss but in most cases, it causes emotional distress in patients because the visible lesion impairs one's self-esteem and social ability (Shahrad and Marks 1976). Several therapies have been used to deal with scalp psoriasis, including corticosteroids and vitamin D3 analogs, keratolytic, dermatome shaving, pulsating magnetic fields, phototherapy, and Grenz rays, use of antifungals, coal tar, Retinoids, and dithranol preparations. All the topical treatments for scalp psoriasis affect the hair condition, such as brittle hairs, oily hairs, and dryness of hairs (Stanley Chan et al. 2009).

Another common involvement of psoriasis is *nails*. Changes in nails can sometimes aid in the diagnosis of psoriasis. The involvement of nails is a particularly aggravating problem for patients because nails are difficult to treat and most of the time respond slowly to any therapy (Salomon et al. 2003). Patients with nail psoriasis are often embarrassed in public, refusing to allow their hands or feet to be seen. The prevalence of nail psoriasis ranges from 10% to 55%. A frequency of around 85% is found in patients with psoriatic arthritis. Only 1% to 3% of patients with nail psoriasis can occur without joint or cutaneous involvement (De Jong et al. 1996). Subungual hyperkeratosis was the most common nail abnormality observed on both toenails and fingernails (79.5% and 91.2%, respectively). The symptoms of nail psoriasis are pitting of the nail plate, transverse ridging, and-splinting hemorrhages, and oil drop discoloration.

The available therapies for nail psoriasis are frequently ineffective, accompanied by unfavorable systemic side effects, time-consuming, or impractical. There are various causes for limited therapeutic effectiveness. Nail psoriasis negatively affects the quality of a patient's life, as it is a highly exposed and apparent disease. Thus, the majority of patients do hope for

a quick recovery. Modern belief, biologics are accessible, however, none of the treatments that are life, currently available can satisfy patient's demands. Due to long-term therapy and/or insufficient convenience of therapeutic realization, particularly for topical treatment, the number of patients is exhibiting decreasing compliance. Patients need to be aware that therapeutic response is assessed in terms of months and sometimes there is no immediate reward (Rich and Scher 2003). It is often difficult to generate significant concentrations of Antipsoriatic active medicines in the affected nail, nail bed, or matrix via topical treatment due to the anatomical nature of the nail unit. The agents suggested for the topical treatment of nail psoriasis are corticosteroids, tazarotene, anthralin, calcipotriol, and 5-fluorouracil.

Intertriginous psoriasis is a kind of psoriasis characterized by inflammatory lesions in skin folds. It is also referred to as flexural or inverse psoriasis. Approximately 2% to 6% of psoriasis patients have the involvement of intertriginous psoriasis. It usually appears in the armpits, groin, under the breasts, and in skin folds around the genitals and buttocks, especially the retro auricular folds. Intertriginous psoriasis patients frequently report acute itching, irritation, sweating, and discomfort, with emotional effects. Patients with inverse psoriasis may have significant comorbidity with diabetes, obesity, and hypertension. In the treatment of intertriginous psoriasis, a special issue needs to be considered: For treatment, it is important to notice improved hydration and percutaneous drug absorption. For example, skin occlusion improves the percutaneous absorption of substances like alcohol, steroids, or phenols. So, the general specifications for treatment plans or their equivalent formulations in intertriginous areas are having no taste or odor, being cosmetically accepted, being chemically and physiologically stable, and not causing irritation. Because it tends to induce irritation, tar preparations, anthralin, and tazarotene do not provide first-line treatment to patients with flexural psoriasis (Gottfried 2008). Table 1 presents the prevalence of psoriasis severity in hard-to-treat areas.

2.4 Risk factors for the development of psoriasis

Psoriasis causes considerable physical, social, and psychological difficulties in addition to its clinical symptoms (Rapp et al. 1999). Psoriasis makes it difficult to sleep, move, use your hands, sit or stand for extended periods, or carry out your job duties. Aghaei et al. (2009) Psoriasis can manifest at any age, psoriasis is likely influenced by a person's ethnicity, genetic history, and environmental factors (Parisi et al. 2013). From the previous survey, the Japanese population observed aggravating factors to be stress (6.4% to 16.6%), illness (3.5% to 8.3%), exposure (1.3% to 3.5%), sun seasonal factors (9.7% to 13.3%), and -blockers (0.9% to 2.3%) (Kawada et al. 2003). The factors also include cardiovascular diseases (4.2% to 8.1%), hypertension (1.1% to 27.8%), tonsillitis (3.5% to 5.4%), and diabetes mellitus (DM) (7.0% to 13.9%) (Ito et al. 2018). Extrinsic and intrinsic risk factors can be used to categorize the risk factors for psoriasis depicted in Fig 3.

External factors of psoriasis include: air pollutants and sun exposure, drugs, vaccination, infection, alcohol, and smoking. *Air Pollutants and Sun Exposure* is the external factor of psoriasis. The increase in air pollution adversely affects the human skin, various air

Table 1 Prevalence of psoriasis severity in hard-to-treat areas

Conditions	Scalp (%)	Face (%)	Palm (%)	Nails (%)
Mild	48.1	27.6	13.5	25.6
Moderate	57.8	41.8	22.9	31.1
Severe	66.1	53.3	19.5	42.4

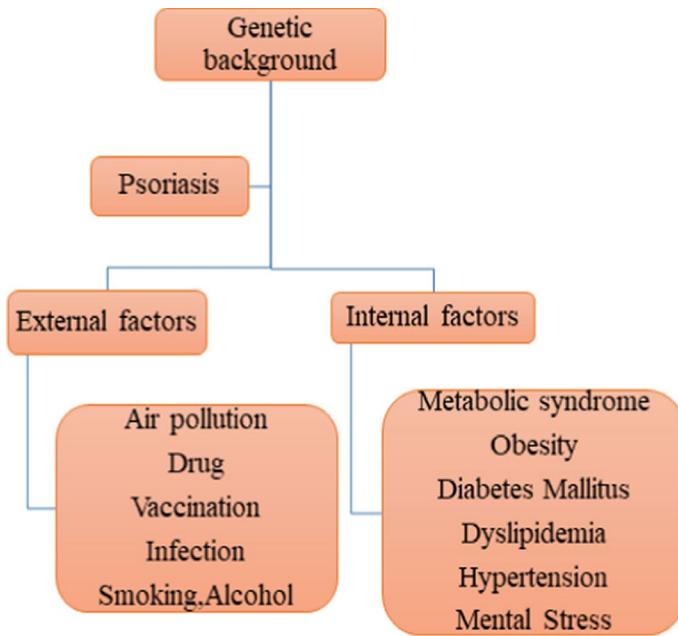


Fig. 3 Taxonomy depicting the risk factors for the development of psoriasis

pollutants such as polycyclic aromatic hydrocarbons, oxides, volatile organic compounds, heavy metals, particulate matter, ozone, and UV can damage the skin. One of the air contaminants that have an impact on the pathogenesis of psoriasis is cadmium. The severe psoriasis patient has higher blood cadmium as compared to the general population. According to this study, environmental cadmium exposure may impair immunity, and microenvironmental disturbances may increase the risk of psoriasis. Two subtypes of UV radiation reach the surface of the earth: 1%–5% UVB (280–315 nm) and more than 95% UVA (315–400 nm). Phototherapy has been used to treat psoriasis extensively for several past decades. Currently, the first-line treatment is a combination of narrowband UVB (311 nm) and excimer laser (308 nm). Psoralen UVA (PUVA) is also used as a second-line treatment for psoriasis, with a preference for thick psoriasis plaques. There is a group of patients with extremely photosensitive psoriasis whose illness is primarily photo distributed and gets worse in the summer.

The worst development of psoriasis is recognized as *Drug-related psoriasis*. In clinical situations, it might be difficult to determine the drug-related causes of psoriasis. This is due to the fact that the latency period between the emergences of psoriasis skin Lesions and the start of medications greatly varies between drugs. Plaque psoriasis, nail psoriasis, palmoplantar psoriasis, pustular psoriasis, scalp psoriasis, and erythrodermic psoriasis are among the psoriasis manifestations caused by drugs.

Another common external factor of Psoriasis is *Vaccination*. Psoriasis patients are increased by a high risk of infection, mostly because of treatment with immunosuppressive or immunomodulatory drugs. In order to avoid the risk of infection vaccination is recommended. Psoriasis can be developed by post-BCG vaccination. BCG has also been used as local immunotherapy for bladder cancer, and a case of erythrodermic pustular psoriasis

induced by BCG immunotherapy has been reported (Jadoon et al. 2023; Attique et al. 2022). Adenovirus vaccination was found to increase the likelihood of psoriasis in a retrospective investigation. Other vaccinations like the tetanus-diphtheria vaccine and the pneumococcal polysaccharide vaccine may also cause psoriasis.

Infection is also an external factor of Psoriasis. Streptococcal infection and psoriasis are known to be associated with one another. Psoriasis develops after streptococcal infection, and guttate psoriasis is the most common type. Despite being self-limiting, the symptoms might return if the streptococcal infection returns. As a result, tonsillectomy may be an option for individuals with persistent psoriasis linked to bouts of tonsillitis (Raharja et al. 2021; Smith and Barker 2006).

Psoriasis has been associated with both *alcohol and smoking*. According to a systematic review and meta-analysis, psoriasis patients are more likely to be current or past smokers. Smoking has been associated with a higher risk of psoriasis. Furthermore, smoking is highly associated with pustular psoriasis lesions. With the increase of pack years or smoking duration, the risk of developing psoriasis has increased. Another investigation revealed a connection between the duration and amount of smoking with the development of psoriasis. It seems that alcohol consumption also appears to be a high-risk factor for developing psoriasis.

Internal factors of psoriasis are Metabolic syndrome, Mental stress, Diabetes Mellitus (DM), Dyslipidemia, and Hypertension. *Metabolic syndrome* is prevalent in psoriasis patients, and *obesity* is highly associated with psoriasis severity. Psoriasis patients have a much greater prevalence and risk of obesity. Obesity is defined as an increase in white adipose tissue, and the mediators produced by adipose tissue that cause low-grade inflammatory processes that lead to the development of psoriasis pathogenesis.

Mental stress develops when people believe that the demands on them exceed their ability to manage. Mental stress is the most common cause of psoriasis patient *Diabetes Mellitus (DM)* prevalence may be comparable across a range of patient groups, ethnic origins, and first treatments. Psoriasis and DM were linked, according to a meta-analysis. The link between psoriasis and the likelihood of developing DM has also been shown by several meta-analyses. DM is divided into two categories: type 2 DM and type 1 DM. Psoriasis patients are far more likely to develop type 2 diabetes. Psoriasis patients have a significantly higher risk of 2 DM type.

Dyslipidemia is more common in psoriasis patients and is probably going to get worse as the condition gets worse. In previous research with 70 psoriasis patients, dyslipidemia was found in 62.85% of the patients. Hypertriglyceridemia (39%) and hypertriglyceridemia with a decreased value of high-density lipoprotein were the most prevalent conditions (HDL). While using oral systemic treatments for psoriasis, dyslipidemia might also manifest. *Hypertension* is also an internal factor of Psoriasis. Psoriasis patients demonstrated a higher frequency and incidence of hypertension in a meta-analysis. Additionally, this meta-analysis found that severe psoriasis was linked to a higher risk of hypertension. Psoriasis patients appear to have hypertension that is more severe. A multicenter no interventional observational research with 2210 psoriasis patients found that 26% of them had hypertension and that this prevalence was greater in psoriasis patients than in the general population. On the other hand, psoriasis incidence may be correlated with hypertension.

3 Overview of artificial intelligence, machine learning and deep learning

Dermatology has taken the lead in the use of artificial intelligence in medicine because of its extensive clinical, dermoscopic, and dermatopathological images. Artificial intelligence is an area of computer science that simulates intelligent human behavior using computer machines and programming (Chan et al. 2020). It is difficult to define artificial intelligence (AI) exactly. Alan Turing offers the well-known Turing test in his landmark article “Computing Machinery and Intelligence,” in which a computer is regarded as intelligent if it is indistinguishable from a person in conversation by an unbiased observer (Du-Harpur et al. 2020). AI can be classified into two categories: strong AI and weak AI. Strong AI, often known as “artificial general intelligence”, grants a computer human-level intellect. This complicated system should be able to self-learn how to do a variety of jobs. Machines with awareness, ethical cognition, and multitasking skills fall into this wide group (Barbieri et al. 2018). We currently only have access to weak or narrow AI, in which a machine learns to execute a specific task. With weak AI, several programs must be designed to fulfill diverse jobs (Hogarty et al. 2019).

The word “Artificial Intelligence (AI)” is now frequently used interchangeably with “Machine Learning (ML)” or “Deep Learning (DL)”. ML is a technology made up of a subset of artificial intelligence that allows the aims of artificial intelligence to be achieved. ML encompasses a wide range of algorithms and statistical methodologies, such as random forest, logistic regression, KNN, SVM and so on (Rehman et al. 2019a). Although ML may appear perplexing at first, it is closely connected to classic statistical models that most dermatologists are familiar with (Chan et al. 2020). Figure 4 presents the Venn diagram of Artificial Intelligence algorithms.

Machine learning aims to train machines using the available methods and data. Machines learn how to make decisions based on the information and data that have been analyzed. ML is dynamic, which means it may change when presented with new information. The ‘learning’ component of ML refers to the algorithms’ efforts to reduce mistakes and increase the likelihood of their predictions (Jakhar and Kaur 2020). Machine learning techniques are broadly categorized into three types: supervised, unsupervised, and reinforcement learning (Rehman et al. 2023). In supervised learning, machines learn to select the correct solution through trial and error. A dataset must be given as inputs (called features) and outputs for Artificial learning (called labels). In supervised learning, a model is represented with a dataset $D = \{x, y\}_{n=1}^N$ of input features x and labels y . Supervised learning finds and predicts the best model parameters Θ based on loss function $L(y, \hat{y})$. The output $\hat{y}$ is obtained by giving input x to the model function $f(x; \Theta)$. The most prevalent kind of learning in dermatology is supervised learning. For example, images of pigmented lesions are referred to as “features” in an algorithm that determines whether a lesion is cancerous or benign (Rehman et al. 2020). The system is initially labeled using labeled images of melanoma and benign pigmented lesions, and then the computer applies this knowledge to an unknown collection of skin scans (Chan et al. 2020). On the other hand, unsupervised learning takes simple inputs (unlabeled data) and can find unknown groupings or abnormalities in data. Reinforcement learning combines supervised and unsupervised learning that learns via error, trial, and environmental input (Murphy 2012).

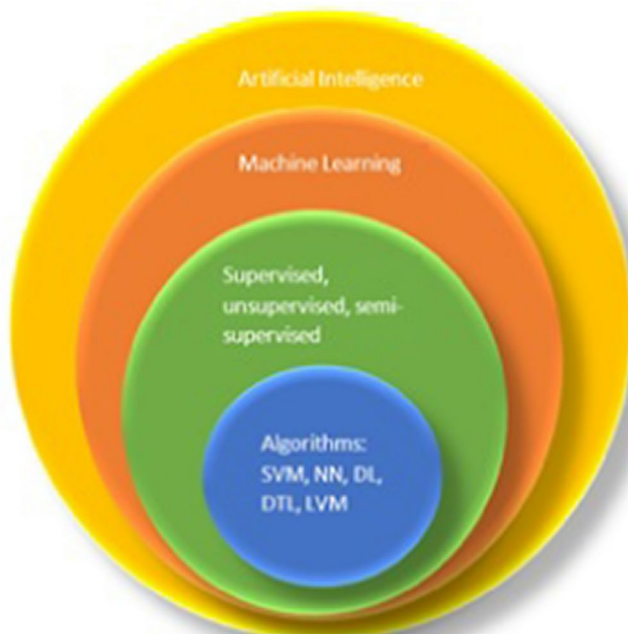


Fig. 4 Venn diagram of artificial intelligence

Several machine learning algorithms are generally used in dermatology. Linear regression, random forest (RF), k-nearest neighbor (k-NN), logistic regression, support vector machine (SVM), and natural language processing (NLP) are some of the most often used statistical learning approaches (Bibi et al. 2020). The number of neighbors in k-NN is utilized for regression and data categorization (Mahesh 2020). Support Vector Machine (SVM) is the widely used conventional machine learning classification model. The training dataset of p points is passed to SVM in the form of $(y_1, z_1) \dots (y_p, z_p)$ where the z_n represents the class to which real vector x_n belongs. SVM soft margin can be computed and minimized as:

$$\left[\frac{1}{p} \sum_{n=1}^p \max(0, 1, z_p(w \cdot y_n - b)) \right] + \lambda \|w\|^2 \quad (1)$$

Hyper plane is represented by w . y_i , value for λ represents that y_i lies on the exact side of margin size.

K Nearest Neighbor (KNN) is a non-parametric conventional machine learning model used for classification. KNN takes the value $\mathbb{R}^d$ from the pairs (Y_1, Z_1) , $(Y_2, Z_2) \dots (Y_n, Z_n)$. The probability distribution P_d is the class label Y with respect to Z , $Y | Z = d \sim P_d$ for $d=1,2$, and p_d . On point y , some norm $\|\cdot\|$ is given on $\mathbb{R}^d$. The training dataset $(Y_{(1)}), (Z_{(1)}) \dots (Y_{(n)}), (Z_{(n)})$ be reordered to $\|Y_{(1)} - y\| \leq \dots \leq \|Y_{(n)} - y\|$. In the training phase of KNN, feature vector and class labels are used. The k-nearest neighbor classifier can be viewed as assigning the k-nearest neighbors a weight of $1/k$ and weights to 0.

Let K_n^{wn} is the weighted nearest neighbor with weights $w_{ni}\}_{i=1}^n$. The asymptotic expansion on class distributions is given as:

$$\mathcal{R}_{\mathcal{R}}(K_n^{wn}) - \mathcal{R}_{\mathcal{R}}(K^{Bayes}) = (C_1 s_n^2 + C_2 t_n^2) \{1 + O(1)\}, \quad (2)$$

where C_1 and C_2 are constants.

$$s_n^2 = \sum_{i=1}^n w_{ni}^2 \text{ and } t_n = n^{-2/d} \sum_{i=1}^n w_{ni} \{i^{1+2/d} - (i-1)^{1+2/d}\}.$$

Random forest is also used for the classification of psoriasis as conventional machine learning approach. The training dataset $(y_i, z_i)\}_{i=1}^n$ makes prediction $\hat{z}$ for new points y' , by looking at the neighborhood of the point, computed by a weight function W :

$$\hat{z} = \sum_{i=1}^n W(y_i, y') z_i \quad (3)$$

$W(y_i, y')$ is the non-negative weight of the i 'th training point relative to the new point in the same tree x' . In tree, $W(y_i, y') = \frac{1}{k'}$, if y_i is one of the k' points in the same leaf as x' , otherwise zero. A forest averages the predictions of a set of m trees with individual weight functions W_j , and its predictions are:

$$\hat{z} = \frac{1}{m} \sum_{j=1}^m \sum_{i=1}^n W_j(y_i, y') z_i = \sum_{i=1}^n \left(\frac{1}{m} \sum_{j=1}^m W_j(y_i, y') \right) z_i. \quad (4)$$

Artificial neural networks (ANNs) or neural networks (NNs) are composed of interconnected units. The network of neurons in the human brain served as an inspiration for ANNs. The ANN's neurons, or nodes, are arranged into linear arrays known as layers. Every node receives input through connections with other nodes that have associated weights. To create an ANN, you must decide on the number of nodes for each layer, the number of levels in the network, and the direction of connections between the nodes. A typical ANN consists of input, output, and hidden layers. A neural network consists of neurons with some activation a and parameters $\Theta = W, B$, with set of weights W and biases B . The activation a is the linear combination of input x to the neuron with element-wise non-linearity transfer function $\sigma(\cdot)$, given as:

$$a = \sigma(w^T x + b) \quad (5)$$

The learning process of machine learning and deep learning is controlled by the explicitly hyper-parameters. The parameters used in the learning process of ML and DL are train-test split ratio, number of hidden layers, number of epochs, number of iterations, kernel size, batch size, pooling size, choice of loss function, choice of activation function, and dropout rate.

Usually, the transfer function in ANN is a sigmoid or hyperbolic tangent. ReLU and tanh are also used as activation function. ANN consists of several layers of transformations followed as:

$$f(x, \Theta) = \sigma(W^T \sigma(W^T \dots \sigma(W^T x + b)) + b) \quad (6)$$

The final layer of ANN uses the softmax function in which activations of the network are mapped to the distribution of classes. The softmax function $P(y|x : \Theta)$ is given as:

$$P(y|x : \Theta) = \text{softmax}(x : \Theta) = \frac{e^{W_i^T x + b_i}}{\sum_{k=1}^K e^{W_k^T x + b_k}} \quad (7)$$

Stochastic gradient descent (SGD) is an iterative method for optimizing an objective function of maximum likelihood. SGD fits the parameters Θ to the dataset $\mathbf{B}$ to minimize the negative log-likelihood in such a way:

$$\arg \min_{\Theta} - \sum_{n=1}^N P[(y_n | x_n : \Theta)] \quad (8)$$

Despite the progress of machine learning in the field of psoriasis recognition, there are several disadvantages of machine learning (Rehman et al. 2021). One of the major limitation is the lack of data. As neural networks require enormous amount of data which is structured to meet the model's requirement for accurate results, however, the amount of data in publicly available psoriasis detection are not enough to train neural networks. Machine learning models depends on the algorithms on which it works. Models do not learn on an incremental basis. Supervised learning is required in machine learning which means human oversight is necessary. Machine learning models are not suited for symbolic, reasoning or high level planing. These limitations are overcome with the emergence of deep learning models. Deep learning models do not require extensive reserves of pre-structured data for results generation. Deep learning models learn on an incremental basis and do not require any human insight. Deep learning models works on symbolic, reasoning or high level planing, speech, vision, playing games and learning natural languages.

Deep learning is part of a broader family of machine learning methods based on artificial neural networks with representation learning (Rehman et al. 2019b). A specific subclass of ANNs known as convolutional Neural Network (CNNs) contains one or more layers, known as convolutional units (pooling units). Two-dimensional or three-dimensional inputs are passed into CNNs and processed through several hidden layers. At each layer, the image is convolved with a set of $\mathbf{K}$ kernels and weights $W = \{W_1, W_2, W_3, \dots, W_K\}$ and biases $B = b_1, b_2, b_3, \dots, b_K$. A feature map X_K is generated with element-wise non-linear transformation function $\sigma(\cdot)$. Each convolutional layer followed the same process with convolutional layer l:

$$X_k^l = \sigma(W_k^{l-1} * X^{l-1} + b_k^{l-1}) \quad (9)$$

Image may be broken down into groups of pixels that serve as the fundamental unit of analysis. Each component of an input image is matched with a small sub-image in the first few CNN layers. Each node is assigned a certain characteristic (such as a shape, color, or size), and the output to the subsequent layer relies on how closely an area of the picture resembles

the allocated feature. This process is carried out through convolution. After these pooling layers, convolution layers, a kind of NN, categorize the entire image (Nichols et al. 2019).

Deep learning and machine learning technologies have shown effectiveness in predicting and categorizing a wide spectrum of diseases (Rehman et al. 2022). Multiple computer approaches are used in deep learning that exhibits the capacity for learning and adaptation. Image segmentation is used to divide the skin images into distinct regions that have same attributes to detect the skin lesion. A deep learning based segmentation model maps the input image x to the segmentation map $\hat{y}$ with parameters θ formalized as a function $\hat{y} = f_{\theta}(x)$. To detect the skin lesion, the binary mask $\hat{y}$ is separated from the skin image. The segmentation model is represented with a dataset $D = \{x, y\}_{n=1}^N$ with training images x_i and ground truth masks y_i such that $(x_i, y_i) : i = 1, 2, \dots, N$. The model parameter θ is maximize by likelihood of observing given data such as:

$$\theta^* = \arg \max_{\theta} \sum_{n=1}^N \log P(y_i | x_i; \theta) \quad (10)$$

Loss function of estimated and true segmented mask is given as the maximization of:

$$\theta^* = \arg \min_{\theta} L(\hat{y} | y_i) = \arg \min_{\theta} L(f_{\theta}(x_i) | y_i) \quad (11)$$

Literature divulges that different deep learning models have been deployed for skin disease detection e.g. Fully Connected Network (FCN) (Attia et al. 2017; Bi et al. 2017; Alenezi et al. 2023), SegNet (Şahin et al. 2022; Naz et al. 2019), UNet (Reddy et al. 2023), YOLO (Aishwarya et al. 2023), and GAN models (Qin et al. 2020).

Ensembling techniques combines different individual algorithms or models to achieve better generalization and accurate performance. Deep ensemble learning models integrates the capabilities of ensembling technique and deep learning techniques in such a way the final model outperforms the generalization. Deep ensemble learning consists of feature extraction using deep learning model, generating base learners, and forming ensemble learner. Let various deep learning extracting methods $E = \{e_1(*; \theta_1^e), e_2(*; \theta_2^e), \dots, e_n(*; \theta_n^e)\}$ where $e_n(*)$ is parameterized by θ_n^e . Features can be extracted from dataset D and extracting methods as:

$$F = \text{feature_extraction}(D, E) = \{f_1, f_2, \dots, f_n\} \quad (12)$$

Dataset D , Targets T , Deep learning algorithm $A = \{dl(*; \theta^{dl}); do(*; *; \theta^{do})\}$ with learning strategy $S = \{ls(\theta^{ls})\}$ are used for generating base learners.

$$BL = \text{base_learner_generator}(D, T, A, S) = \{dl_1(*; \theta_u^{dl_1}), dl_2(*; \theta_u^{dl_2}), \dots, dl_n(*; \theta_u^{dl_n})\} \quad (13)$$

The ensemble deep learning formation is formed with ensemble criteria $C = \{dl_e(*; \theta^{dl_e})\}$ and predictions of the base learner $P = \{p_1, p_2, \dots, p_n\}$. $dl_e(*; \theta^{dl_e})$ represents ensemble deep learning $dl_e(*)$ parameterized by θ^{dl_e} . The ensemble deep learning formation is formed as:

$$EDLF = ensemble_deep_learner_formation(P, T, C) = \{dl_e(*; \theta_u^{dl_e})\} \quad (14)$$

Machine learning and deep learning based ensembling models used for the detection of psoriasis are stacking (Verma and Pal 2020), bagging (Moon et al. 2021), boosting (Terui et al. 2000), implicit or explicit ensembles, negative correlation based deep ensemble models, heterogeneous or homogeneous ensemble, and fusion ensemble technique (Chatterjee 2018).

Deep learning and machine learning technologies have been employed in many medical disciplines for diagnostic reasons (Aijaz et al. 2022). In dermatology, a variety of deep-learning techniques have been applied to accurately anticipate and categorize skin issues. Skin analysis algorithms have been created using the Mask RCNN, transfer learning, and CNN frameworks for classifying skin pictures for the diagnosis of skin lesions, such as malignant melanoma, basal cell carcinoma, actinic keratosis, squamous cell carcinoma, and psoriasis (Moon et al. 2021).

4 Research methodology

In order to explain the research methodology of the review paper, we used PRISMA methodology. The PRISMA statement outlines a standardized approach for reporting systematic reviews and meta-analyses. This framework enables reviewers to clearly articulate the review's objectives, methods, and results. The main objectives of research methodology are:

- A detailed explanation of the introduction has been provided
- A comprehensive background has been provided, which thoroughly explains the types of psoriasis, pathogenesis, affected areas, and risk factors
- A detailed overview of public and private datasets has been presented, including sample images and tables that provide a detailed analysis of the samples in each dataset
- Aims to review, evaluate, and synthesize the concepts of machine learning and deep learning in psoriasis recognition systems
- An in-depth examination of the challenges, limitations, and future prospects has been conducted, providing a detailed explanation to aid new researchers and promote further research on psoriasis

The PRISMA methodology outlines the main steps of the review methodology, which include information sources, selection criteria, and search and selection procedures. *Information Sources* is the initial step in the systematic research methodology involved a meticulous literature search to retrieve relevant articles. Google Scholar was utilized as the primary search platform, and a comprehensive scan of references was performed to present a thorough and authoritative review. The second step is the *Selection Criteria* that entailed a systematic selection process, applying the following inclusion–exclusion criteria to the retrieved literature are: Studies published as articles or reviews. Studies written in the English language. Studies focused on psoriasis recognition systems. Studies published between 2017 and 2024. *Search and selection procedure* is the last step that involved a systematic search of studies from diverse information sources, employing a predefined set of keywords pertinent to psoriasis recognition systems. The publications are searched in the information

sources by using the keywords “psoriasis”, “psoriasis detection using machine learning”, “psoriasis detection using deep learning”, “dermatology”, and “skin disorders”. A total of 150 publications are included in this survey paper. The PRISMA methodology of the review paper is depicted from Fig. 5.

5 Dataset

For the diagnosis system of psoriasis detection and dermatological diseases, the first step is to prepare the dataset. The dataset is the foundation of any research project that can be acquired from different sources like specialized hospitals and clinics, approved repositories, official websites, and electronic sites over the Internet. After reviewing extensive review of psoriasis-related literature, we observed that the majority of research is conducted in partnership with hospitals, and consequently, the datasets are private. Furthermore, complementing the private datasets, there exist publicly available psoriasis datasets that offer valuable resources for research. In this section we discussed both private and publicly available datasets.

Shrivastava et al. (2015) collected 40 skin images (270 psoriasis and 270 non-psoriasis) from 30 Indian psoriasis patients. Hasija et al. (2017) conducted an experiment on 3000 images sub-divided into four classes: Eczema, Psoriasis Herpes, and Melanoma. In pre-processing stage they performed gray scaling, blurring sharpening, increasing the contrast, changing the color channel, reducing the noise, and smoothing imbalanced dataset issues. Zhang et al. (2018) employed inception v3 and Google net on a clinical database that contains 28,000 dermoscopic images. Zhao et al. (2020) collected 28,653 dermoscopic images of different skin diseases from the Dermatology Department, Peking Union Medical College Hospital. Wu et al. (2020) conducted an experiment on 4,740 images collected from the Dermatology Department of Central South University and Xiangya Hospital. In data preparation, they cleaned images by removing duplicate cases and blur images. Yang et al. (2021) deployed EfficientNet-B4 architecture on 7033 dermoscopic images. They collected images from 1166 patients from Union Medical College Hospital of China. They performed

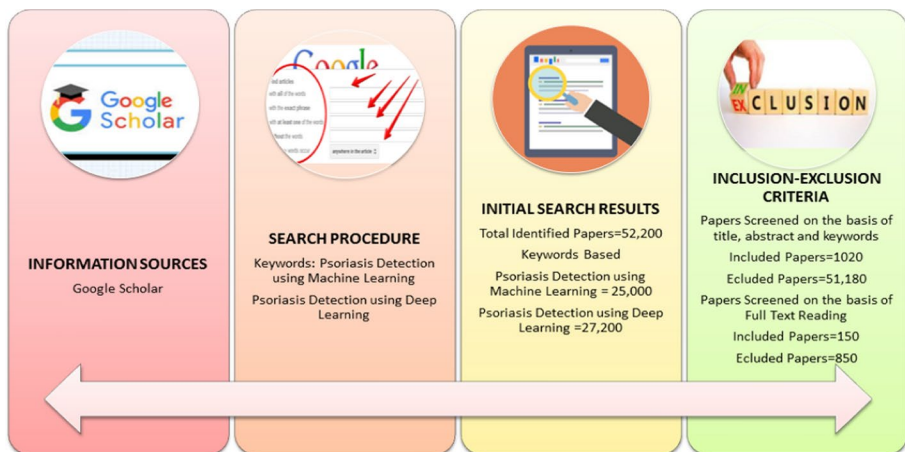


Fig. 5 PRISMA methodology of psoriasis detection system

rescaling, cropping, and horizontal and vertical flipping in preprocessing. Chakraborty et al. (2024) used four deep transfer learning models: VGG16, VGG19, MobileNetV1, and ResNet-50 for the recognition of psoriasis. They collected 352 images from local hospital and applied preprocessing techniques to generate 496 image patches.

After discussing the private datasets, now, we will delve into the details of publicly accessible datasets. Publicly accessible datasets refer to repositories of data that are freely available for utilization by researchers, developers, and other stakeholders. These datasets are hosted online and can be downloaded without any restrictive licensing agreements. The publicly available datasets for psoriasis detection are discussed below:

Dermnet Dataset (Dermnet) includes 301 total images that are related to five types of psoriasis. The Dermnet dataset comprises 23 various types of skin disorders, such as plaque, pustular, guttate, erythrodermic, and inverse psoriasis. Alopecia areata, eczema, and poison are the other types of skin diseases included in this dataset. A few samples of the Dermnet dataset have been depicted in Fig. 6.

Dermatology Dataset (Dermatology dataset) contains 33 linear valued attributes and 1 nominal attribute. Dermatology has a serious challenge with the differential diagnosis of erythematous-squamous disorders. The dataset includes different types of skin disorders like seborrheic dermatitis, Psoriasis, chronic dermatitis, pityriasis rosea, pityriasis rubra pilaris, and lichen planus. Some samples of the dermatology dataset have been shown in Fig. 7.

AtlasDerm is commonly referred as Atlas of Dermoscopy dataset (Argenziano et al. 2000, Atlasderm dermatology diseases). Its main purpose was to help clinicians in the identification of dermoscopic signs associated with melanoma and the diagnosis of skin lesions. Each image is watermarked and simply carries the label of disease diagnosis in this dataset. The dataset comprises 130 psoriasis images, 175 lichen planus, 53 eczemas, 275 melanocytic nevus images, 30 vascular skin lesions images, and 582 melanoma images. Few sample images of AtlasDerm are shown in Fig. 8.

DermIS (Dermis) is commonly known as the Dermoscopic dataset Dermatology Information System. This dataset was developed through collaboration between the University of Erlangen's Department of Dermatology and the University of Heidelberg's Department



Fig. 6 Samples of dermnet dataset



Fig. 7 Samples of dermatology dataset



Fig. 8 Samples of AtlasDerm dataset

of Clinical Social Medicine. The dataset consists of 6588 images in which there are 239 psoriasis, 212 lichen planus, 244 eczema, 243 BCC, and other skin disease images. This dataset includes conventional disease diagnostic labels as well as text descriptions of lesion location, race, and age for each image. The disadvantage of this dataset is that there are not many images in each class, and the images are watermarked. Pediatric dermatology online image atlas (PeDOIA) a dermatology and online image atlas (DOIA) divide this dataset into two sections. About 600 dermatological diagnoses are covered by the DOIA's 3000 images of lesions. Dermoscopic images, preliminary and differential diagnoses, case reports, and



Fig. 9 Samples of DermIS dataset



Fig. 10 Samples of DermQuest dataset

other data on essentially each type of skin disorder are included. Few sample images of DermIS dataset are depicted in Fig. 9.

Derm Quest is publicly available and comprised of 22,082 dermoscopic images. Only the DermQuest dataset contains lesion labels. For each image in the dataset, there were around 134 labels for lesions. In 2018, the DermQuest dataset was transferred to *Derm101*. However, on December 31, 2019, this dataset was disabled. Derm101 contains 470 psoriasis, 60 lichen planus, 3474 BCC, 806 eczema, and other skin diseases. Figure 10 shows sample images of DermQuest Dataset.

Dander is a data collection website for skin disease images from Denmark. There are a total of 1110 images and 91 different types of skin diseases. The dataset contains 43 psoriasis, 17 lichen planus, and the rest of the images belong to other skin disease images. The

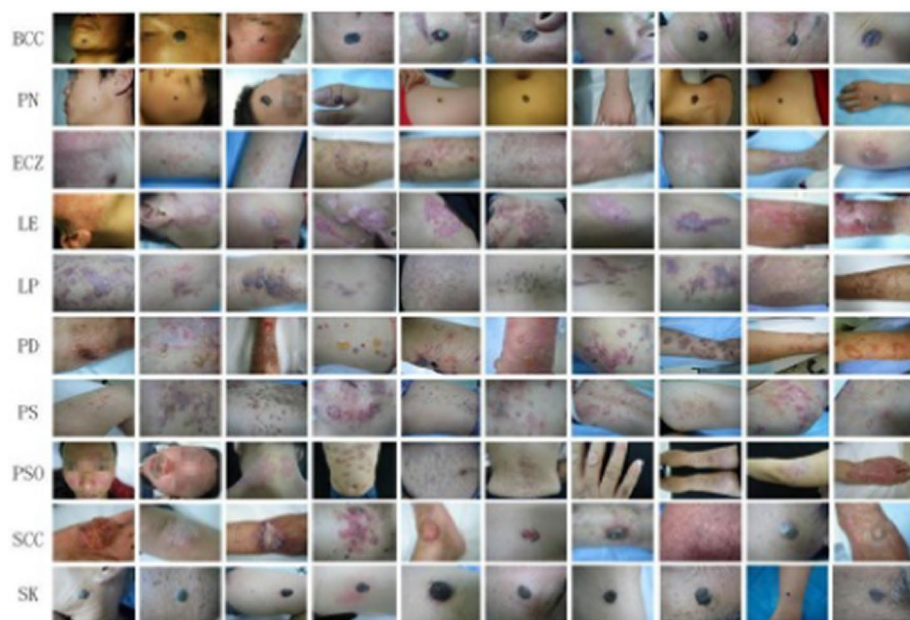


Fig. 11 Samples of XiangyaDerm dataset

Table 2 Publicly available dermatological datasets

Dataset	Classes	Amount	No of samples in each class label								
			Pso	LP	Par	LE	EcZ	BCC	SCC	Pem	SK
Dermnet	626	18,974	985	271	–	207	993	539	135	120	398
AtlasDerm (2000)	534	9,503	130	175	15	141	53	277	121	118	11
DermIS	735	7,172	239	212	9	160	244	243	103	134	48
Derm101	525	22,979	470	60	–	59	806	3474	310	13	1220
XiangyaDerm-Pso9 (2019)	107,565	541	900	900	900	900	900	832	900	900	889

majority of the patients in this dataset are watermarking occlusion and have simply the label of disease diagnosis.

The Ethics Committee of Xiangya Hospital of Central South University authorized the collection of *XiangyaDerm* Xie et al. (2019). It comprises 107,565 clinical images covering 541 different forms of skin disorders. Each image in this collection has been labeled by a competent doctor. It is the largest clinical images dataset of Asian skin disorders utilized in Computer Aided Diagnosis (CAD) systems globally.

Samples of XiangyaDerm dataset are depicted in Fig. 11. The details of the publicly dermatological datasets according to disease classes, images amount, and diseases labels are explained in Table 2.

6 State of art

Psoriasis is an insistent, relapsing, chronic, and inflammatory skin disorder with no treatment and cannot be passed from one individual to another. According to the IFPA (International Federation of Psoriasis Associations) report, approximately 125 million people are affected by psoriasis globally and 3 to 5% of Pakistani people are suffering from psoriasis (George et al. 2016). Although psoriasis is not curable but with some effort, it can be well-managed. Artificial Intelligence (AI) may serve as a tool in clinical assessment and help develop personalized treatment protocols and outcome predictions. Computer-based intelligence programs for detecting, monitoring, and diagnosing psoriasis are as yet being refined. The early diagnosis of psoriasis intensely depends on the visual perception of the skin lesion biopsy by an experienced dermatologist (Greaves and Weinstein 1995). This diagnosis procedure is inaccurate, less effective, and labor-intensive. In 2014, Guo et al. (2014) developed the first AI method that identifies psoriasis. They worked on two microarray-based gene expression datasets: GSE14905 and GSE 13355. They applied three featured selection algorithms on 21 psoriasis-associated features, which were then used to create the classifier. The resultant binary classifier achieved an average overall accuracy of 99.81%. After that system, different machine learning and deep learning techniques were deployed in the domain of psoriasis. This section presents the notable contributions of conventional machine learning and deep learning techniques for psoriasis detection and classification.

6.1 Conventional machine learning models based classification

Machine learning (ML) is a technology made up of a subset of artificial intelligence that allows the aims of artificial intelligence to be achieved. ML encompasses a wide range of conventional algorithms and statistical methodologies, such as random forest, logistic regression, support vector machine, KNN, etc.

Liu et al. (2025) conducted experiment on psoriasis datasets GSE30999, GSE109248, GSE14905, and GSE201827 downloaded from the Gene Expression Omnibus (GEO) database. Machine learning models were employed to predict psoriasis onset and validated using external datasets.

Sun et al. [2025] performed integrated analysis of five psoriasis microarray datasets (GSE30999, GSE106992, GSE14905, GSE78097, and GSE117468) to identify immune-key genes. Weighted Gene Co-expression Network Analysis (WGCNA) and differential gene analysis revealed six immune-related hub genes (CLEC7A, CXCL1, IRF1, S100A12, S100A8, and S100A9) with high predictive accuracy (AUC values: 0.956–0.989).

AlDera and Othman (2022) proposed a system that provides an easy, fast, and efficient solution for skin disease diagnoses such as acne, cherry angioma, melanoma, and psoriasis using image processing and machine learning techniques. They conducted experiment on DermNet NZ and the atlas dermatological dataset containing 377 images of four different disease classes: acne, cherry angioma, melanoma, and psoriasis. They resized the images to 250×250 pixels and removed noise. Furthermore, the color images were converted to grayscale in preprocessing. By using machine learning algorithms like Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbor (K-NN) classifiers they attained 84.2%, 90.7%, and 67.1%, respectively. Aijaz et al. (2022) extracted machine-learning features like color, texture, and shape and passed to CNN and LSTM for classification.

Naveen Raju et al. (2021) proposed a support vector machine based on black widow optimization for skin disease classification. They conducted experiment on the ISIC-2018 dataset which contains images of five different skin diseases, including benign with healthy images, melanoma, psoriasis, herpes, and paederus. They eliminated noise from the input images in preprocessing. The enhanced fuzzy set segmentation approach was then used to segment the skin infection region. Features such as shape, gray-level co-occurrence matrix texture, and color texture were extracted from the segmented region. They attained 92% accuracy in their experimentation.

Tapak et al. (2021) organized two machine-learning models of the genetic algorithm and support vector machine (SVM) for the classification of gene expression signatures for psoriasis. They conducted experiment on the GSE55201 dataset. They extracted 181 hand-crafted features and achieved 100% accuracy. Anandhi and Wiselin (2021) worked on 731 images based on six categories of psoriasis that are taken from a Vivitar 7.1 Megapixel Digital Camera. The main objective of the work was to develop a system that estimates the parameter of disease and predicts the type of psoriasis. They employed a Genetic Algorithm (GA) and Multiclass Particle Swarm Optimization (MPSO) that classifies the types of psoriasis disease using the selected features. They achieved an accuracy of 95.7% from the proposed method.

ALenezi (2019) conducted work on the custom dataset by collecting images from different websites specific to skin diseases. The dataset consists of 80 images of every disease i.e. 20 Melanoma images, 20 Normal images, 20 Eczema images, and 20 Psoriasis images. They used the SVM and convolutional neural network AlexNet for the detection of skin diseases and the system successfully detects 3 different types of skin diseases with 100% accuracy.

Ubale and Paikrao (2019) proposed a color phase model for identifying and classifying various skin disorders. Using the HSV and the LAB color phase model, they extracted features from the images. In pre-processing, they performed image resizing, RGB to HSV, lab conversion, and contrast enhancement. Finally, they classified skin diseases (Acne, Papillomas, Psoriasis, Melanoma, Mycosis, Vitiligo, and Warts) using the K Nearest Neighbor (KNN) classification approach. They attained an accuracy of 91.80% for the HSV Color Phase Model and an accuracy of 81.60% for the LAB Color Phase Model. Moreover, the preprocessing was not good enough, and their work has no authenticated dataset. Guo et al. (2014) proposed an AI algorithm that aimed to identify psoriasis. The proposed system employed microarray-based gene expression on two datasets: GSE14905 and GSE 13355. They applied three featured selection algorithms on 21 psoriasis-associated features, which were then used to create the classifier. The resultant binary classifier achieved an average overall accuracy of 99.81% over three independent validation strategies.

Pal et al. (2018) conducted experiment on 90 images. They performed image cropping in preprocessing and computed handcrafted features like color and texture features. They fed the extracted features to classification algorithms like KNN, SVM, and RF. They achieved an accuracy of 88.58% using KNN, 92.17% using SVM, 92.14% using RF respectively. Jarad and Dawood (2018) conducted work on 200 color images of psoriasis in which there are 100 normal and 100 abnormal cases with 200×200 pixels. They used color and texture features and employed machine learning using an Artificial Neural Network classifier ANN to obtain optimized performance. They attained 100% accuracy in this approach.

Patrick et al. (2018) used genetic data to assess the risk of psoriatic arthritis development in psoriasis patients. They conducted experiment on six cohorts with ≥ 7000 genotyped psoriatic arthritis and cutaneous-only psoriasis patients. They used five GWAS datasets CASP19, Kiel21, Exomechip with GWAS content20, Genizon21, PsA GWAS13, and Immunochip dataset. They employed machine learning approaches to other types of omics data, to classify psoriasis. Random forests have been used to predict psoriasis from transcriptome data and electronic records and support vector machines have been used for dermoscopic images to predict psoriasis.

Shrivastava et al. (2015) developed an automated method to risk stratify psoriasis severity from healthy and diseased skin images using color feature patterns. They collected 40 skin images (270 healthy and 270 diseased) from 30 Indian psoriasis patients. They employed SVM, and PCA and obtain 99% accuracy. Arifin et al. (2012) proposed an automated and robust method for the diagnosis of different dermatological diseases. The system operates in two phases, the first detecting skin irregularities and the second diagnosing dermatological. The system employs color k-means clustering, image processing techniques, and color gradient techniques to identify skin diseases. They conducted an experiment on 2055 images in which 704 are 6 skin diseases images such as (Eczema, Acne, psoriasis, Scabies, Tinea Corporis, and Vitiligo). In pre-processing they segment the healthy skin and diseased one by making the system for classification using cropping and color gradient generation. The system exhibits a skin disease detection accuracy of 95.99% and a disease identification accuracy of 94.016%. Hameed et al. (2020) proposed a unique Computer-aided Diagnosis (CAD) method for identifying the most common skin lesions such as acne, eczema, psoriasis, and benign and malignant melanoma. They used the Dull Razor approach, Gaussian filtering for preprocessing, and Otsu's thresholding method to segment the desired region from which the features are extracted. Several color and texture features were extracted and used in the quadratic kernel Support Vector Machine (SVM). They conducted an experiment on 15,546 images collected from different sources and achieved 83% accuracy for six-class classification.

Sinthura et al. (2020) conducted an experiment on a small dataset containing 100 images. They proposed an advanced skin disease diagnostic model based on image processing. Firstly, they used Otsu's approach to segment the disease part. Following that, certain GLCM properties were extracted, including area, mean, perimeter, and entropy. Finally, the SVM classifier was used to detect four types of diseases: acne, melanoma, psoriasis, and rosacea. They achieved 89% accuracy in identifying skin disorders. Gunathilake (2021) conducted an experiment on the classification and identification of skin diseases using machine learning techniques like a neural network, convolutional neural network, K-means clustering, support vector machine used to solve the problems of the skin disease diagnosis method. They used color-based models, Gray Level Co-Occurrence Matrix, and YCbCr algorithms for the feature extraction of the image. For image processing techniques they used Sobel operators and edge detection. The data set is collected from the most appropriate and accurate websites. They attain 80–100% accuracy from machine learning-based classification. The review of psoriasis detection systems using machine learning techniques are summarized in Table 3.

Table 3 Review of psoriasis detection systems using machine learning

References	Dataset	Methods/approaches	Performance %
Aldera and Othman (2022)	DermNet NZ and the atlas dermatological dataset containing 377 images	Resized to 250X250, removing noise, converting color images into grayscale. Extracted color features and passed to Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbor (K-NN) for classification	84.2, 90.7, 67.1
Ajiaz et al. (2022)	BFL NTU dataset, Dermnet dataset	Segmentation, data augmentation, and enhancement. Extracted color, texture, and shape features. LSTM is used for classification	84.2 and 72.3
Naveen Raju et al. (2021)	ISIC-2018	Noise reduction. Extracted Color and texture features. SVM is used for classification	92
Tapak et al. (2021)	GSE55201 dataset	Dimensionality reduction and feature selection. Genetic Algorithm and SVM are used for classification	100
Anandhi and Wiselin (2021)	731 images based on six categories of psoriasis	Segmentation, feature extraction, morphological closing method, and Indexing. Extracted color and texture features. Genetic Algorithm (GA) and Multiclass Particle Swarm Optimization (MPSO) is used for classification	95.7
Hameed et al. (2020)	15,546 images collected from different sources	Dull Razor approach, Gaussian filtering. Extracted color, and texture. Support vector machine (SVM) is used to classify skin lesions	83
Sinhura et al. (2020)	100 images of skin diseases	Otsu's approach to segment the diseased part. Extracted Model-based features. SVM is used for classification	89
ALenezi (2019)	80 images collected from different websites of skin diseases	Feature extraction and classification. Extracted Model based features. Alex net and SVM used for classification	100
Jarad and Dawood (2018)	200 color images	Extracted color and texture features. Machine learning using Artificial Neural Network classifier ANN to obtain optimized performance	100 form CADP
Shrivastava et al. (2015)	40 skin images (270 healthy and 270 diseased) from 30 Indian psoriasis patients	Extracted color features. SVM and PCA are used for classification	99
Guo et al. (2014)	GSE 13555 and GSE14905	Extracted Model-based features. Novel Incremental Feature Selection algorithm is used for classification	99.81
Arifin et al. (2012)	2055 images of skin diseases	Cropping and Color Gradient Generation. Extracted model-based features. Color k-means clustering, image processing techniques, and color gradient techniques to identify skin diseases	Skin disease detection accuracy: 95.99, Disease identification accuracy: 94.016

6.2 Deep learning models based classification

Deep learning is the broader family of machine learning methods based on artificial neural networks with representation learning. In this section, we present some of the notable contributions of deep learning for the detection of psoriasis.

Eskandari and Sharbatdar (2024) conducted a comprehensive study of deep learning applications in skin disease diagnosis, specifically focusing on psoriasis and lichen planus. They employed advanced data augmentation and class balancing techniques to enhance dataset diversity and mitigate class imbalance. They implemented a custom 50-layer ResNet-50 architecture to classify these diseases, yielding an impressive accuracy of 89.07%.

Yaseliyani et al. (2024) developed an ensemble Convolutional Neural Network (CNN) model for detecting and classifying psoriatic images, utilizing ResNet50V2, ResNet101V2, and ResNet152V2 networks. The model determines whether an image is psoriatic and classifies it into one of seven variants, followed by personalized treatment recommendations using Multi-Classification Decision Making. A web-based Decision and Diagnostic Support System (D&DSS) is developed to facilitate accurate diagnosis and effective treatment planning for psoriasis patients. They achieved the highest accuracy of 91.90% using ensemble technique for binary classification and 93.29% on Multi-Classification respectively.

Müller et al. (2024) developed and validated a diagnostic algorithm to identify the “Flip-Flop” (FF) phenomenon in patients with atopic dermatitis (AD) and psoriasis vulgaris (PV). The algorithm combined clinical criteria and was prospectively evaluated using the CK-CARE registry. Machine learning (ML) enhancements improved the algorithm’s performance, yielding a Youden index of 78.9% and accuracy of 89.7%. This study presents the first validated tool for identifying FF phenomenon, offering a valuable diagnostic aid for clinicians.

Xiao et al. (2024) developed an automated diagnosis system for psoriasis using deep learning (DL). The system extracts morphological and cellular features from whole slide images (WSI) to detect psoriatic cells. A novel Convolutional Neural Network (CNN) architecture balances classification accuracy and model complexity. The proposed method involves patch-level classification, heat map generation, and machine learning-based feature extraction for WSI-level classification. The results demonstrated high precision in the diagnosis of psoriasis, achieving 92.90% at the patch level and 98.18% at the WSI level.

Chakraborty et al. (2024) used four deep transfer learning models: VGG16, VGG19, MobileNetV1, and ResNet-50 for the recognition of psoriasis. They collected 352 images from local hospital and applied preprocessing techniques to generate 496 image patches. The highest accuracy of 99.13% is obtained using MobileNetV1 with Adaptive gradient optimizer.

Rashid et al. (2024) proposed a light-weighted deep neural network (DNN) for the detection of Psoriasis. They convert the RGB images into YCbCr in preprocessing for accurate classification. They reported the accuracy of 99.89%, AUC of 0.99, sensitivity of 99.0%, and specificity of 99.08% respectively.

Lee et al. (2023) developed the prediction model for psoriatic arthritis. They collected the dataset from Taiwan’s National Health Insurance Research Database and split it into the training set of 80% and test set of 20%. They employed 8 layers CNN architecture for classification and attained AUC of 0.70, sensitivity of 0.80, and specificity of 0.60 respectively.

Aijaz et al. (2022) employ CNN and LSTM for the classification of types of psoriasis. They conducted an experiment on 172 images taken from the BFL NTU dataset and 301 images of psoriasis taken from the Dermnet dataset. They performed segmentation, data augmentation, and enhancement which was followed by color, texture, and shape feature extraction. The accuracy attained from the CNN model is 84.2% and 72.3% from LSTM respectively.

Sari et al. (2023) conducted an experiment on a dataset obtained from DermIs, DermNet, and DermatoWeb. Dermatitis atopic and psoriasis skin disease images are included in the dataset. They used 2640 images in the training set, 661 in the testing set, and 75 in the unseen data. The resolution of the image was 224 X 224. The augmentation includes flipping, shear, zooming, and rotation of images. They proposed 10 CNN models to achieve the best outcome for classifying psoriasis and dermatitis atopic. They used VGG 16, Mobile Net, VGG 19, DenseNet 121, ResNet 50, ResNet 101, Mobile Net V2, DenseNet 201, Inception, and Xception. The best results are achieved from the inception V3 model with 84% accuracy for data testing, 82% accuracy for unseen data and confusion matrix with a True positive achieved is 248, True Negative is 61, False Negative of 298, and false positive is 54. Hsieh et al. (2022) conducted an experiment on nail psoriasis detection. They employed mask RCNN for the segmentation of 200–300 images of training sets and 100–150 images for validation sets. They reported mean accuracy of 91.5% for each class.

Alhamdani and Hashim (2022) proposed a system to classify psoriasis and eczema using a convolution neural network (CNN). They conducted an experiment on 5564 images that are collected from the Dermnet and Kaggle websites. From the proposed system they achieved an average classification accuracy of 96.13%. Arunkumar and Jayanna (2022) proposed a system by using depth-wise separable convolution neural networks to classify psoriasis-affected skin into one of four types: inverse, erythrodermic, guttate, nail, plaque, and pustular. They conducted an experiment on 6400 images collected from the Department of Dermatology, Navodaya Medical College, Raichur, and Rani Channamma University, Belagavi. During the preprocessing stage, the input images were segmented into 224 pixels X 224 pixels using the Keras Image Data Generator function. They used MobileNet machine learning architecture which is based on depth-wise separable convolutions and attained 86% classification accuracy.

Various deep learning architectures have been explored for segmenting psoriasis lesions from full-body color images. Kovvuru et al. (2025) investigated the use of overcomplete U-Net variants, including Residual U-Net and Attention U-Net, to improve the accuracy of lesion segmentation. The results showed that these variants outperformed the standard U-Net, with the Overcomplete Attention U-Net achieving the highest performance (Dice similarity index: 0.9834). This suggests that overcomplete U-Net variants are promising architectures for objective and accurate segmentation of psoriasis lesions.

Pal et al. (2018) employed a U-shaped fully convolutional Neural Network (FCN) for the segmentation of Psoriasis skin biopsy images. They conducted an experiment on 90 images. They performed image cropping in preprocessing and labeled the images using super-pixel generation, semantic labeling, and smoothing the super-pixel class labels. They also computed handcrafted features like color and texture features and then fed these features to classification algorithms like KNN, SVM, and RF. They achieved an accuracy of 88.58% using KNN, 92.17% using SVM, 92.14% using RF, and 95.17% on FCN respectively. Schaap et al. (2022) conducted an experiment on 5844 images collected from the Department of

Dermatology at the Radboud university medical center, Nijmegen, the Netherlands. They aimed to evaluate and compare the performance of CNNs to image-based automated PASI scoring in anatomical areas. They trained CNNs using on 614 arms, 576 trunks, and 541 leg regions of the imaging series. CNNs were trained independently for the PASI subscore of erythema, desquamation, induration, and area in each anatomical location (trunk, arms, and legs). The erythema, desquamation, induration, and area CNN and real-life scores for the trunk region were 0.616, 0.580, 0.580, and 0.793, respectively, with comparable findings for the arms and legs region. Wibowo et al. (2022) conducted an experiment on 1203 images consisting of different types of skin diseases including psoriasis. They proposed a system to detect common skin diseases using ensemble learning and CNN methods. They achieved 93% accuracy on Grayscale images which are superior to RGB images producing only 86%. Ramo (2022) conducted an experiment on 250 images. They aimed to develop a computer intelligent system that identifies and diagnoses the type of skin inflammation. They proposed SANN (Snake Algorithm Neural Network) system and attained 88.9% accuracy.

Amin and Farooq (2021) conducted an experiment on 800 images, comprising 300 images of psoriasis taken from DermNetNZ and Google. They performed image augmentation in preprocessing like horizontal flip, vertical flip, random resized crop, and random rotation. They applied different transfer learning models for the classification of psoriasis detection. The highest accuracy of 94% was achieved on a test set for the classification of two classes: psoriasis and healthy. Yang et al. (2021) deployed EfficientNet-B4 architecture on 7033 dermoscopic images. They collected images from 1166 patients from Union Medical College Hospital of China. They performed rescaling, cropping, and horizontal and vertical flipping in preprocessing. They reported a sensitivity of 92.7% and specificity of 82.7% using two classes (psoriasis and healthy), and a sensitivity of 88.9% and specificity of 96.8% using four classes (psoriasis, eczema, lichen planus, and others respectively).

Mustafina and Sukhov (2020) conducted an experiment on 7000 dermoscopic images of diseases like Melanoma, Nevus, Seborrheic Keratosis, psoriasis, etc. They proposed a web-based application to diagnose skin diseases using deep learning techniques such as CNN architecture and three predefined models known as ResNet 152V2, AlexNet, and InceptionV3. They attained 99% accuracy using CNN, 88.83% using ResNet 152V2, 65.46% with Inception, and 74.89% using AlexNet in the training set. Hridoy et al. (2021) conducted an experiment on 52,500 images of skin diseases and healthy skin. They used horizontal and vertical flips, Rotation transformations (including 90, 180, and 270 degrees), random zoom augmentation, and horizontal and vertical shift, techniques. They proposed a system to diagnose twenty different types of skin diseases using eight architectures of Efficient Net between B0 and B7. They have trained using the transfer learning approach. They achieved the highest accuracy from EfficientNet-B7 i.e. 97.10% and 93.35% accuracy from EfficientNet-B0 respectively.

Kadry et al. (2021) implemented Convolutional Neural Network (CNN) based approach for the identification and extraction of skin melanoma. They conducted an experiment on 1250 numbers of RGB-scaled images taken from the DDI dataset. Using the VGG-SegNet scheme, they computed 97.16% accuracy.

Roslan et al. (2020) conducted an experiment on IPC and DermNetNZ in which there are 187 total images, 82 images for plaque psoriasis and 105 images for guttate psoriasis. In pre-processing, they resize the image into 160×160 pixels to achieve precise classifica-

tion. They used CNN for image preprocessing and achieve 82.9% and 72.4% for Plaque and Guttate Psoriasis skin disease.

Sinkov et al. (2016) conducted an experiment on the XiangyaDerm dataset called XiangyaDerm Pso9, which contains 8021 photographs of psoriasis with 8 other dermatological diseases. The images were resized to 300×300 pixels in preprocessing. They developed an algorithm to detect the skin infection on the desired area, and also, to distinguish a specific skin disorder from others, taking into account the texture features and color characteristics of the skin to improve the accuracy of detecting the disease. They deployed four deep CNN architectures: DenseNet, Inception V3, Inception ResNetV2, and Xception and reported a classification accuracy of 87% respectively. Goceri and Karakas (2020) conducted an experiment on different skin disease images to classify psoriasis and other skin disorder. They use deep learning neural network models such as VGG16, VGG19, Google Net, InceptionV3 and, ResNet101. The evolution of the result from this model is based on accuracy. From the result, ResNet101 has the highest accuracy which is 77.72%. Wu et al. (2020) proposed an artificial intelligence dermatology diagnosis assistant (AIDDA) for psoriasis, eczema AD, and healthy skin based on the EfficientNet-b4 CNN algorithm. They conducted an experiment on 4,740 images collected from the Dermatology Department of Central South University and Xiangya Hospital. In data preparation, they cleaned images by removing duplicate cases and blur images. AIDDA has an overall diagnostic accuracy of $95.80 \pm 0.09\%$, sensitivity of $94.40 \pm 0.12\%$, and a specificity of $97.20 \pm 0.06\%$. The accuracy for Psoriasis is 89.46%, with a sensitivity of 91.4% and a specificity of 95.48%, while the accuracy for AD and Eczema is 92.57%, with a sensitivity of 94.56% and a specificity of 94.41%, according to AIDDA.

Roshni Thanka et al. (2020) proposed a system using a deep convolutional neural network to classify the different skin lesions. They conducted an experiment on the DermNet dataset in which there are 1800 images of nine different skin diseases including psoriasis. Finally, they attained 96% accuracy for individual models amongst the set of models. Zhao et al. (2020) conducted an experiment on the XiangyaDerm-Pso9 dataset containing 8021 images of nine common diseases. They Developed a CNN to classify skin images of 9 common skin disorders (lichen planus, lupus erythematosus, basal cell carcinoma, squamous cell carcinoma, eczema, pemphigus, psoriasis, and Seborrheic keratosis) as psoriasis vs. nonpsoriasis. They deployed different deep learning models including, including Inception-ResNetV2, DenseNet121, Xception, and InceptionV3 to evaluate the results. Goceri and Karakas (2020) conducted experiments on different skin disease images to classify psoriasis and other skin disorder. They use deep learning neural network models such as VGG16, VGG19, Google Net, InceptionV3, and, ResNet101. The evolution of the result from this model is based on accuracy. From the result, ResNet101 has achieved the highest accuracy which is 77.72%. Padilla et al. (2019) conducted an experiment on 6,264 images for training from the Dermatology atlas: AtlasDerm, Danderm, DermIs, DermNet, DermQuest, and DermatoWeb. They used MobileNet architecture and attained 88% and 90% respectively in classifying Atopic Dermatitis and Psoriasis Chronic Plaque. Połap et al. (2018) Combined artificial intelligence and sensors to develop a smart home system for evaluating skin disorders. They used the PH2 dataset with normal and melanocytic skin lesion images and performed image resolution in preprocessing. They deployed CNN architecture and attained 82.4% accuracy.

Zhang et al. (2018) employed inception v3 and Google net on a clinical database that contains 28,000 dermoscopic images. They divide the data set into two classes Dataset A and Dataset B. They conducted an experiment to classify the four skin diseases Seborrheic keratosis (SK), basal cell carcinoma (BCC), melanocytic nevus, and psoriasis. They perform Hair/Line, Air Bubbles/ Dermoscopic Gel, and Brightness Unbalanced in preprocessing that are misclassified causes under the “Interference Factor” category. The accuracy achieved by the deep learning algorithms is $87.25 \pm 2.24\%$ from Dataset A (1067 images) and $86.63\% \pm 5.78\%$ from dataset B (528 equally graphs distributed) which is promising. Rathod et al. (2018) conducted an experiment on the DermNet dataset to classify and diagnose various kinds of skin diseases. They converted images into the standard size of 120X120, rotated at 90 degrees, and also flipped them. From convolutional neural network techniques, high-level features such as edge, border, and color are obtained using ConvNet such as Max Pooling, Convolution, etc. The initial training gives an accuracy of 70% approximately and the large data set can increase the accuracy to more than 90 percent.

Munia Tamanna et al. (2017) conducted an experiment on a dataset consisting of 45 psoriasis images. During the pre-processing phase, these images were cropped and unwanted noise was removed so that skin and psoriasis region remained. They resized the images 154×151 to 1800×1700 pixels using color features. They aimed to propose a novel region-based color clustering segmentation for recognizing psoriasis lesions automatically from images. The suggested modified k-means clustering-based segmentation method achieved a 93.83% accuracy. With an acceptable accuracy of 86.83%, the classification model SVM predicted the disease plaques, and therefore the automated psoriasis segmentation and plaque localization approach was developed. Hasija et al. (2017) conducted an experiment on 3000 images sub-divided into four classes: Eczema, Psoriasis Herpes, and Melanoma. In pre-processing stage they performed gray scaling, blurring sharpening, increasing the contrast, changing the color channel, reducing the noise, and smoothing imbalanced dataset issues. They used multiple AI algorithms like Convolutional Neural Networks and Support Vector Machines and combined them with image processing tools to develop a better structure, leading to higher accuracy of 95.3%. Islam et al. (2017) conducted an experiment on Dermnet Skin disease dataset. They proposed multiclass skin disease detection system based on image texture. Various pre-processing techniques were applied such as contrast enhancement, resize, grayscale conversion, and noise removal. The GLCM technique was used to extract the image textures, and Maximum Entropy Thresholding was used to segment the images. Finally, three separate classes of skin disease images-Eczema, Impetigo, and Psoriasis-were classified using the Back propagation (BPN) method. The accuracy achieved for this method was reported at 80% with sensitivity and specificity of 71.4% and 87.5%, respectively. Pal et al. (2016) conducted an experiment on a new annotated dataset containing 707 images. In the pre-processing stage, images are first normalized 227×227 before passing to the network. They aimed to predict severity scores using novel deep CNN-based architecture. The experimental findings reveal that deep CNN-based architectures (both STL and MTL) yield promising results, and MTL achieves higher accuracy of 85.29%. The review of psoriasis detection systems using deep learning techniques are summarized in Table 4.

Table 4 Review of psoriasis detection systems using deep learning

References	Dataset	Methods/approaches	Performance
Sari et al. (2023)	DermIs, DermiNet, and DermatoWeb	Flipping, shear, zooming, and rotation of images. Extracted Model based features. VGG 16, Mobile Net, VGG 19, DenseNet 121, ResNet 50, ResNet 101, Mobile Net V2, DenseNet 201, Inception, and Xception were applied	84 from inception V3
Alhamdani and Hashim (2022)	Derm Net	Extracted Model-based features. CNN is used for classification	96.13
Arunkumar and Jayanna (2022)	6400 images of skin diseases	Images are segmented into 224 pixels X 224 pixels using the Keras Image Data Generator function. Extracted Model-based features. MobileNet is used for classification	86
Schaap et al. (2022)	5844 images of skin diseases	Extracted Model-based features. CNN architecture is used for classification	0.6160, 0.580, 0.580, and 0.793
Wibowo et al. (2022)	1203 images of skin diseases	Extracted Model-based features. Ensemble learning and CNN methods are used for classification	93
Ramo (2022)	50 images of skin diseases	Extracted Model-based features. SANN is used for classification	88.9
Hsieh et al. (2022)	200–300 images of training sets and 100–150 images for validation sets	Extracted Model-based features. Mask RCNN is used for classification	91.5
Amin and Farooq (2021)	DermNet and Google images	Image Augmentation (random rotation, random resized crop, horizontal flip, and vertical flip). Extracted Model-based features. ResNeXt is used for classification	94
Yang et al. (2021)	7033 dermoscopic images	Rescaling, cropping, horizontal and vertical flipping. Extracted Model-based features. EfficientNet-B4 is used for classification	96.8
Mustafina and Sukhov (2020)	7000 dermoscopic images	Extracted Model based features. CNN, ResNet 152V2, Alex Net, and InceptionV3 are used for classification	CNN: 99, ResNet 152V2: 88.83, INCEPTION V3: 65.46, ALEXNET: 74.89
Hridoy et al. (2021)	52500 images of skin diseases and healthy skin	Horizontal and vertical flips, Rotation transformations (including 90, 180, and 270 degrees), random zoom augmentation, and horizontal and vertical shift, techniques. Extracted Model based features. EfficientNet B0 and B7 are used for classification	EfficientNet-B7: 97.10, EfficientNet-B0:93.35
Kadry et al. (2021)	1250 numbers of RGB scaled images taken from DDI dataset	Extracted Model based features. VGG-SegNet is used for classification	97.16
Khan et al. (2021)	ISB12016, ISB12017, HAM1000 datasets	Contrast stretching scheme. Extracted Model based features. DenseNet deep model is used for classification	96.3, 94.8, and 88.5

Table 4 (continued)

References	Dataset	Methods/approaches	Performance
Roslan et al. (2020)	IPC and dermatnet	Resize image into 160 x 160. Extracted Model based features. CNN is used for classification	82.9 and 72.4 for Plaque and Guttate Psoriasis
Mustafina and Sukhov (2020)	DermLine website	Resized to 300x300. Extracted model-based features. Four deep CNN architectures: DenseNet, Inception V3, InceptionResNetV2, and Xception are used for classification	87
Wu et al. (2020)	4740 images collected from the Dermatology Department	Cleaned images by removing duplicate cases and blur images. Extracted Model based features. EfficientNet-b4 is used for classification	95.800.09
Roshni Thanka et al. (2020)	DermNet dataset	Extracted Model based features. Deep convolutional neural network to classify the different skin lesions	96
Goceri and Karakas (2020)	Different skin disease images	Extracted Model-based features. Deep learning neural network models such as VGG16, VGG19, Google Net, InceptionV3 and, ResNet101 were applied	77.72 from ResNet101
Padilla et al. (2019)	6264 images for training from the Dermatology atlas: AtlasDerm, DanderM, DermIs, DermNet, DermQuest, and DermatoWeb. Extracted Model based features	MobileNet	88 and 90
Zhao et al. (2020)	28,653 dermoscopic images of different skin diseases from the Dermatology Department, Peking Union Medical College Hospital	Extracted Model-based features. Residential network is designed to diagnose six types of lesions using transfer learning	65.8 and 90 accuracy on two classification tasks and six-classification
Polap et al. (2018)	PH2	Image resolution. Extracted Model based features. CNN architecture is used for classification	82.4
Zhang et al. (2018)	28,000 dermoscopic images	Hair/Line, Air Bubbles/ Dermoscopic Gel and Brightness Unbalanced. Extracted Model based features. Inception v3 and Google are used for classification	87.2524 86.635.78
Rathod et al. (2018)	Dermnet	Converted images into the standard size of 120X120, rotated at 90 degrees, and also flipped them. Extracted Model based features. Convolutional Neural Network (CNN) is used for classification	70 in training and 90 in large datasets
Hasija et al. (2017)	3000 images	gray scaling, blurring sharpening, increasing the contrast, changing the color channel, reducing the noise, and smoothing. Extracted model-based features. Convolutional Neural Networks and Support Vector Machines are used for classification	95.3

Table 4 (continued)

References	Dataset	Methods/approaches	Performance
Islam et al. (2017)	Dermnet	Contrast enhancement, resize, grayscale conversion, and noise removal. Extracted Model-based features. GLCM technique is used for classification	80

7 Challenges or limitations and future outlooks

Introducing machine learning and deep learning technology into dermatological clinical practices faces both challenges specific to psoriasis and common to dermatology. A significant challenge is data standardization, particularly for skin images, which are frequently taken without a standardized protocol, resulting in variations in color normalization, exposure, perspective, and other parameters. Psoriasis datasets are often small, imbalanced, and scarce. Psoriasis can be difficult to distinguish from other skin conditions, such as eczema, dermatitis, or seborrheic dermatitis. These factors make the difficulty for machine learning algorithms to distinguish between actual and artificial variances between captured lesions. The International Skin Imaging Collaboration has tried to solve this challenge by developing a set of technical guidelines for skin lesion images, but achieving acceptance of these standards across dermatological practices is problematic, especially when images are generated by patients (Katragadda et al. 2017).

A small data set can induce bias and errors, especially in highly expensive systems like deep neural networks, which need a large training set to generate generalized output. Accurate annotation of psoriasis lesions requires expertise, which can be time-consuming and expensive. Deep learning models can be prone to overfitting, especially when trained on small datasets, which can limit their generalizability to new cases. Deep learning models can be difficult to interpret, making it challenging to understand the decisions made by the model. Psoriasis datasets often suffer from class imbalance, where one class (e.g., non-psoriasis) has a significantly larger number of instances than the other class (e.g., psoriasis). Real-world images can vary greatly in terms of lighting, resolution, and other factors, which can affect the performance of deep learning models.

The primary limitation of existing psoriasis disease risk assessment methods is the lack of risk stratification with varying severity levels. Another major challenge is the absence of the adaption of risk assessment tools for multi-center clinical trials that are capable of handling big databases in an automated manner. Rehman et al. (2022); Chen et al. (2018) The third key challenge is determining how to evaluate risk assessment systems utilizing machine learning paradigms employing patient pictures that are a combination of color and grayscale at the global level Shrivastava et al. (2015).

Computer-aided diagnosis systems (CADx) for psoriasis detection also have several notable challenges, such as: (i) Machine learning approaches are affected by acquisition techniques. If the circumstances of image acquisition are changed, re-training is required. (ii) The features must be selected for optimal performance. (iii) Feature selection is a required component of the CADx system. When the feature count is high, dimensionality reduction techniques such as PCA are required. (iv) The optimal training data size must be optimized for the greatest performance and generalization of machine learning systems. (v) The system's performance is sensitive to the adopted partitioning protocol, as well as the types of kernels used during the machine learning system's training phase (Shrivastava et al. 2016).

There are some other challenges related to psoriasis features. Psoriasis can appear in a variety of ways, including plaque, pustular, guttate, nail, and palmoplantar psoriasis. A machine learning algorithm trained on the most common form of psoriasis would not be able to identify unusual appearances as the same disease. Similarly, psoriasis may appear on any part of the body and can vary in form and size within the same patient. Most images capture only psoriasis patches, making it impossible to calculate an exact body surface area

scoring (BSA). ML algorithm may compare lesions on the extremities with those on the body parts. Whole-body imaging is one way to address this issue, but it is not available in most situations, including telemedicine and normal clinic visits (Savolainen et al. 1997).

Even when we focus on particular psoriasis lesions, there are still psoriasis feature characteristics that cause challenges. Erythema, scale, and induration are the three most important factors in determining the severity of a psoriasis lesion. Color normalization and controlled lighting circumstances would help an accurate assessment of erythema; otherwise, a bright red lesion might seem dark brown. The silvery scale of psoriasis is complex in how it reflects light, making it difficult to interpret scaliness. Lastly, induration is difficult to adequately quantify in two dimensions (Adamson and Smith 2018).

To address the challenges associated with the application of deep learning and machine learning in psoriasis, we propose novel solutions and outline the future potentials of ML development in this field. The challenge of training machine learning (ML) models, especially deep learning (DL) models, with limited data is a current research focus. Efforts to collect and annotate large-scale psoriasis datasets would help to improve model robustness and generalizability. Research into new architectures and techniques, such as transfer learning and few-shot learning, could help to improve model performance on small and imbalanced datasets. Folle et al. (2023) demonstrated the feasibility of training models with minimal data for psoriasis diagnosis using the BEiT model. Few-shot learning is a ML paradigm that facilitates efficient model training with limited data. Xiao et al. (2023); Liu et al. (2022); Bolin Chen and Han, and Lin Yan. (2023); Xiao et al. (2023) successfully applied few-shot learning to classify and segment lesion data using a limited number of images. The collection and labeling of psoriasis data are complicated by subjective factors, highlighting the need for innovative solutions. In the current data-driven landscape, smaller, more granular datasets may offer improved performance compared to larger, more extensive datasets.

Inter-image variability can significantly impact model performance, particularly in feature extraction (Maghsoodi and Quincey 2024; He et al. 2024). To address this issue, it is essential to standardize images prior to model training. This can be achieved by extracting common features from images using techniques such as segmentation. Incorporating multiple data modalities, such as images, clinical features, and genomic data, could help to improve model accuracy and provide a more comprehensive understanding of psoriasis. Díaz (2017) employed a segmentation model to pixelate images, thereby reducing inter-image differences by extracting pixel-level features. However, this approach requires labeling, which can be a significant limitation. To address this issue, Pathak et al. (2015) introduced the concept of weak segmentation, which enables automatic segmentation without prior labeling. This approach facilitates the recognition of common attributes between images, thereby enhancing model performance in assessing psoriasis. Furthermore, pre-processing features of skin lesions, such as color, can also be beneficial.

Artificial Intelligence holds significant promise to further improve and develop psoriasis recognition, evaluation, management, treatment, and diagnostics. To aid in diagnosing psoriasis, AI can be used for identifying the psoriasis area from the image, differentiating psoriasis features from other skin disorders, and scoring and detecting the severity of psoriasis affected area. Thus AI provides practical implications in clinical practices of dermatology.

As a treatment and therapy management aid, ML can assist to avoid disease difficulties (Goessinger et al. 2024). For instance, a psoriasis patient anticipated by ML to have qualities putting them at higher risk of cardiovascular complications, as studied by Munger

et al. (2020), could be focused on for protection cardiology administrations. Psoriasis treatment can also be further improved by Artificial Intelligence. Automated lesion-evaluation technologies could help dermatologists in observing and monitoring patients and making treatment decisions. For instance, a high PGA computed by ML could alert a dermatologist that a psoriasis patient may require more intensive systemic treatment or phototherapy over topical. Long-term treatment reactions, drug connections, and possible new treatments for psoriasis can likewise be anticipated by utilizing artificial intelligence techniques.

One potential research direction is the development of multimodal learning approaches that integrate clinical features and medical images to improve psoriasis diagnosis accuracy. This can involve exploring different multimodal fusion techniques, such as early fusion, late fusion, and hybrid fusion, to combine information from different data modalities. Additionally, attention-based models can be developed to focus on the most relevant regions of medical images.

Another area of research is the exploration of advanced deep learning architectures, such as graph convolutional networks and transformers, to model the relationships between different regions of medical images. These architectures can potentially improve the accuracy of psoriasis diagnosis by capturing complex patterns in the data. Furthermore, techniques such as explainable AI and visual explanations can be developed to provide insights into the decisions made by deep learning models.

Real-world deployment of deep learning models for psoriasis recognition is also an important area of research. This involves collaborating with clinicians to validate the performance of models in real-world clinical settings, as well as exploring deployment on edge devices and integration with electronic health records. Moreover, data augmentation and generation techniques, such as generative adversarial networks and 3D printing, can be investigated to artificially increase the size of psoriasis datasets.

Finally, transfer learning and domain adaptation techniques can be explored to adapt pre-trained deep learning models to psoriasis recognition tasks, as well as to adapt models trained on one dataset to another dataset with different characteristics. Few-shot learning techniques can also be investigated to enable deep learning models to learn from a small number of examples.

8 Evaluation, synthesis, and main findings

The survey paper aims to review, evaluate, and synthesize the concepts of artificial intelligence in psoriasis recognition systems, examining both its current applications and its future potential. The survey presents the notable contributions of machine learning and deep learning in the domain of psoriasis detection from 2019 to 2024. The literature review is based on peer-reviewed publications of Google Scholar. The inclusion–exclusion criteria include publications based on articles, conferences, and reviews written in the English language. The publications are searched in the information sources by using the keywords “psoriasis”, “psoriasis detection using machine learning”, “psoriasis detection using deep learning”, “dermatology”, and “skin disorders”. A total of 120 publications are included in this survey paper. Figure 12 shows the number of published articles on psoriasis recognition using ML/DL from 2019 to 2024 across different academic journals. The graph demonstrates a notable peak in 2024, reflecting a sharp rise in researchers’ interest in this field in recent years. This

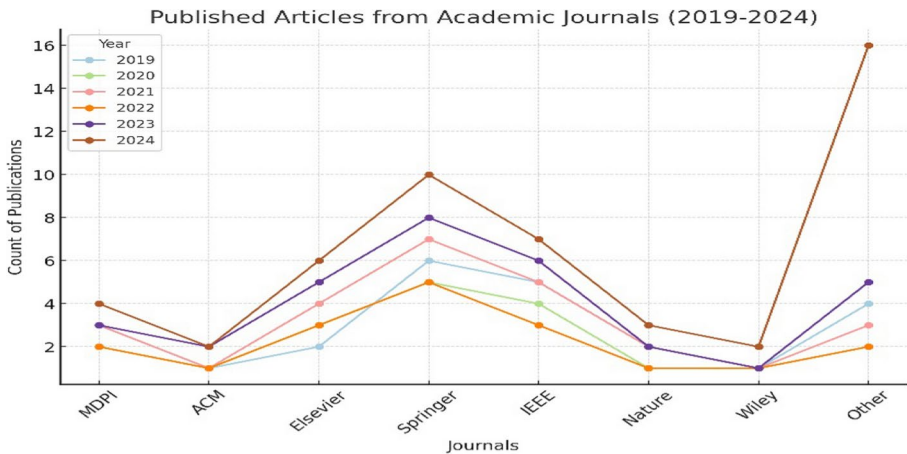


Fig. 12 Published articles from academic journals based on academic journal publication databases and years

surge in published articles highlights the substantial growth in research and development, emphasizing the field's increasing significance within the scientific community. Additionally, the figure examines the distribution of publications across various publishers, revealing that Springer has played a crucial role in disseminating scientific research. Its contributions have been particularly impactful in advancing knowledge compared to other publishers.

A critical analysis of the review article shows that most of the publications used the Dermnet dataset for the detection of psoriasis using both machine and deep learning. Different machine learning algorithms like SVM, KNN, Random Forest, and Genetic algorithms for the classification of psoriasis. Support Vector Machine (SVM) is the widely used conventional machine learning classification model. 80% of the publications used SVM for the classification of psoriasis.

Deep learning is the broader family of machine learning methods based on artificial neural networks with representation learning. In recent years, Convolutional Neural Network (CNN) has been introduced into the detection and classification of dermoscopic images. The CNN uses convolutional filters or kernels to detect the pixel-level descriptive features from a large data set of dermoscopic images and the corresponding diagnosis as input for “end-to-end” training. The strength of CNN lies in its ability to learn “what to look for” without manual guidance. Since dermoscopic images provide high-resolution information and the diagnosis of psoriasis is mainly based on the appearance of the skin lesion, a CNN trained on dermoscopic images would achieve excellent performance in identifying psoriasis, which might improve the management of psoriasis in areas with limited medical resources. However, current studies about the computer-aided diagnosis of skin diseases focus mainly on the classification of psoriasis. A tremendous improvement in the psoriasis recognition system can be observed with the emergence of deep learning.

Transfer learning is the technique of deep learning in which the deep neural network is trained on one base task and utilize the learned knowledge to another target task. Different CNN architectures like AlexNet, GoogLeNet, VGG, ResNet, and DenseNet are used for the transfer learning in the domain of psoriasis detection. Figure 13 presents the statistics

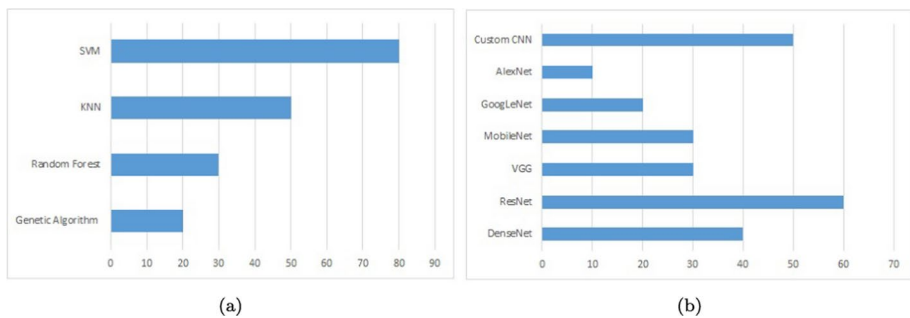


Fig. 13 Statistics of current trends **a** Machine learning algorithms, **b** Deep learning algorithms

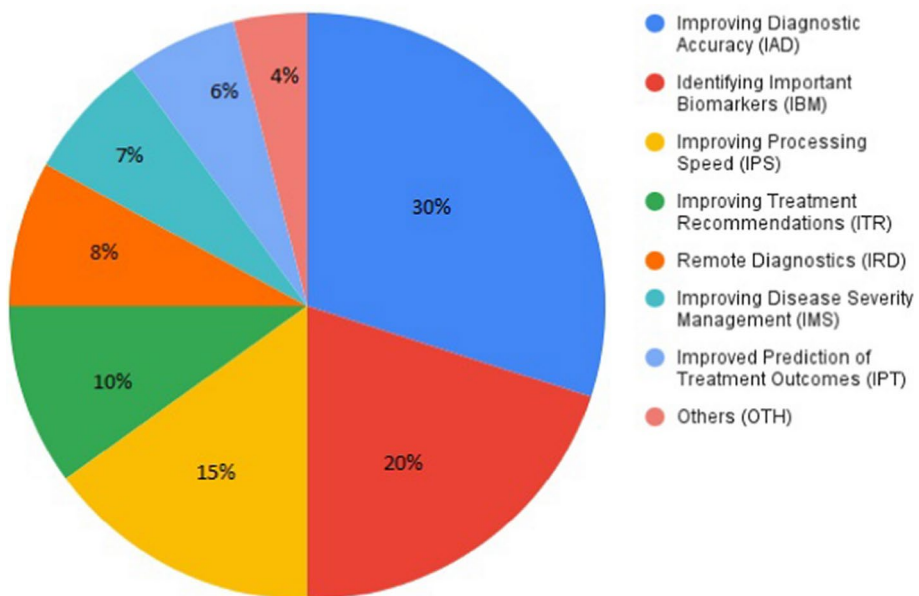


Fig. 14 Comparative analysis of the research advantages of psoriasis

of the current trends of machine learning and deep learning algorithms. It can be synthesized that SVM is the widely used machine learning algorithm for psoriasis detection. Similarly, mostly researchers designed custom CNN deep learning models. ResNet is widely used as a transfer learning deep learning model for the detection of psoriasis respectively.

Figure 14 illustrates a comparative analysis of the research benefits derived from machine learning (ML) approaches in psoriasis, categorized according to the proposed framework. The classification consists of seven key groups: Improving Diagnostic Accuracy (IAD), Identifying Important Biomarkers (IBM), Improving Processing Speed (IPS), Enhancing Treatment Recommendations (ITR), Remote Diagnostics (IRD), Improving Disease Severity Management (IMS), Improved Prediction of Treatment Outcomes (IPT), and Others (OTH). The analysis reveals the following distribution of research benefits: IAD at 30%, IBM at 20%, IPS at 15%, ITR at 10%, IRD at 8%, IMS at 7%, IPT at 6%, and OTH at 4%.

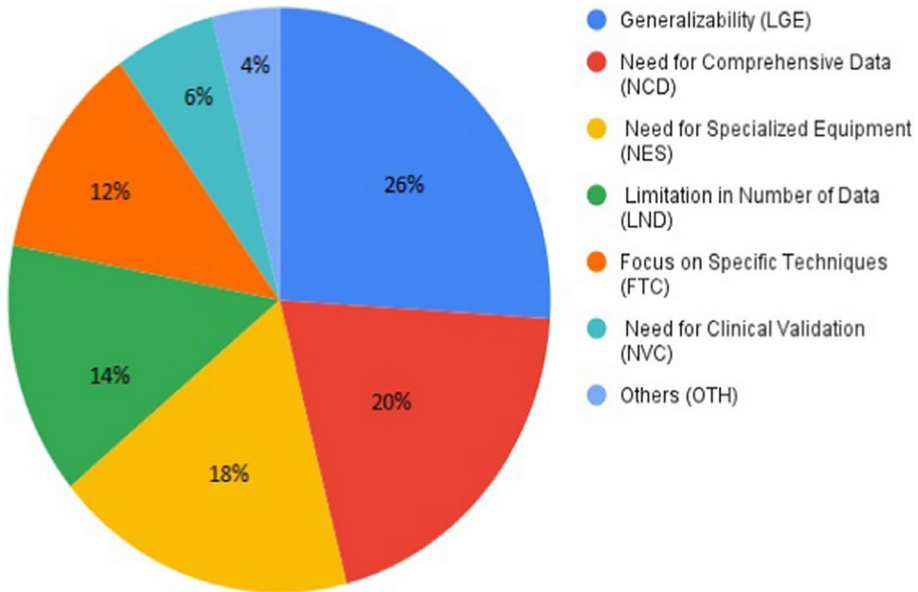


Fig. 15 Comparative analysis of the research disadvantages of psoriasis

In particular, IAD stands out as the benefit most frequently identified, accounting for 30% of the total research efforts.

Figure 15 presents a comparative analysis of the research disadvantages associated with machine learning (ML) approaches in psoriasis, categorized according to the proposed framework. This classification includes six primary groups: Limitation of Generalizability (LGE), Need for Comprehensive Data (NCD), Need for Specialized Equipment (NES), Limitation in Number of Data (LND), Focus on Specific Techniques (FTC), Need for Clinical Validation (NVC), and Others (OTH). According to the analysis, the distribution of research disadvantages (limitations and challenges) for each category is as follows: LGE at 26%, NCD at 20%, NES at 18%, LND at 14%, FTC at 12%, NVC at 6%, and OTH at 4%. Notably, the most significant disadvantage identified is related to LGE, which accounts for 26% of all studies.

9 Conclusion

The integration of AI in dermatological practice is catalyzing a paradigmatic shift in the field, enabling enhanced precision and customization of medical treatments, particularly for patients with psoriasis. This advancement is especially vital in areas with limited access to specialized dermatological care, where AI-driven solutions can help mitigate healthcare disparities. Artificial intelligence (AI) has the potential to revolutionize the treatment of psoriasis, affecting various stages of the disease, from initial diagnosis and risk stratification to personalized treatment recommendations and patient education. The purpose of this survey was to systematically evaluate, synthesize, and analyze the existing literature on AI-based psoriasis recognition systems, with a focus on both current applications and future potential.

We have surveyed 150 publications of psoriasis using machine learning and deep learning. A critical analysis of the review article shows that most of the publications used the Dermnet dataset for the detection of psoriasis using both machine and deep learning. It can be synthesized that SVM is the widely used machine learning algorithm for psoriasis detection. Similarly, mostly researchers designed custom CNN deep learning models. ResNet is widely used as a transfer learning deep learning model for the detection of psoriasis respectively.

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